

COMPUTEUR VISION FOR MARKETING VALUATION

QUANTIFYNG NBA SPONSOR EXPOSURE WITH YOLO-OBB MODELS

Kylian Santos

Table of Contents

1. Introduction and Theoretical Framework	4
1.1. Economic context	4
1.1.1. Key sponsorship figures in the NBA	4
1.1.2. A league focused on highlights	5
1.2. Project Objective	5
1.2.1. The meeting of computer vision and advertising	5
1.2.2. Existing solution	6
2. Data Collection and Processing	8
2.1. Origin and nature of the data	8
2.1.1. Collecting images: Video extraction pipeline	8
2.1.2. NBA API	9
2.2. Labeling Methodology	9
2.2.1. Definition of zone taxonomy	9
2.2.2. Human-in-the-Loop Labeling	10
2.2.3. Iterative Data Enrichment	11
2.3. Exploratory Data Analysis	11
3. Visual Detection Architecture	15
3.1. Model Selection	15
3.2. Training Strategy and Fine-Tuning	16
3.3. Performance Evaluation and Error Analysis	18
3.3.1. Metrics Analysis	18
3.3.2. Error Analysis	20
3.3.3. Interpretability	22
4. Econometric Modeling of Media Value	24
4.1. Use of the model	24
4.1.1. Quantity of exposure	24
4.1.2. Quality of exposure	24
4.2. Financial valuation formula	25
4.2.1. Simple formula	25

4.2.2.	Towards advanced modeling: critical discussion	33
5.	Quantification and Prediction of Media Value.....	37
5.1.	Descriptive Analysis	37
5.2.	Predictive Modeling	44
5.2.1.	Linear Regression	45
5.2.2.	Random Forest and XGBoost	47
5.2.3.	Conclusion and Model Choice.....	50
6.	Conclusion.....	51
6.1.	Synthesis.....	51
6.2.	Biases and Limitations of the Current Mode.....	51
6.2.1.	The model	51
6.2.2.	The data.....	52
6.2.3.	The Valuation Formula	53
6.3.	Model Generalization	53
6.3.1.	In-League Extension	53
6.3.2.	Out-League Extension	54
7.	References	55
7.1.	Market Analysis and Specialized Press.....	55
7.2.	Advertising Costs (CPV) and Sponsorship Data	55
7.3.	Recommendations	55
8.	Appendices	56
8.1.	Methodology for the collection of URLs of interest	56
8.2.	Justification for the Choice of 5 FPS Sampling.....	57
8.3.	Note on the Use of LLMs.....	59

1. Introduction and Theoretical Framework

1.1. Economic context

The National Basketball Association (NBA) is no longer defined solely as a sports league, but as a global financial ecosystem where fan attention is the primary currency. NBA franchises should no longer be seen as teams whose only goal is sports performance, but rather as companies seeking to generate as much revenue as possible. It is noted that today, a franchise is worth an average of 5.5 billion dollars, representing a total valuation of 165 billion dollars for the league's 30 teams. Understanding this economic structure of sponsorship is an essential prerequisite for evaluating the relevance of our valuation approach.

1.1.1. Key sponsorship figures in the NBA

Forty years ago, at the start of Michael Jordan's career, all NBA teams collectively generated 33 million dollars. This amount, while significant at the time, is now negligible compared to the 1.4 billion dollars reached during the 2022/2023 season. This growth continues thanks to the expansion of the league, notably through the NBA Cup and development in Europe. Revenues then increased by 7% to reach 1.5 billion dollars during Victor Wembanyama's rookie season, marked by record audiences, before crossing the 1.7 billion dollar mark for the 2024/2025 season. In total, the sponsorship sector has seen a staggering evolution of 91% over the last five years.

This financial dynamic is fueled by a sophisticated hierarchy of advertising assets, ranging from 6-square-centimeter jersey patches to arena naming rights, the latter representing about 10% of the category. The financial services sector dominates investments with 247 million dollars annually, followed by technology (122 million) and healthcare (105 million). The physical geography of the court is thus segmented into zones of varying economic value, where naming rights stand at the top. Iconic deals, such as the one for the Crypto.com Arena valued at 700 million dollars over 20 years, generally include the most prominent asset on the floor: the central logo (Mid-Court Logo).

These multi-year partnerships, often costing hundreds of millions of dollars, are justified by almost permanent television exposure. Setting these prices relies on a complex equation integrating sports performance, market size, and media exposure metrics on the Web and Digital platforms. This dependency highlights the critical need for granular and precise measurement tools, especially since these revenues represent 14% of team resources

1.1.2. A league focused on highlights

The NBA's digital landscape is distinguished by a unique phenomenon in professional sports: the massive proliferation of unauthorized highlights generated by fans. This ecosystem, which we refer to as "Shadow Media," generates at least hundreds of millions of views each year. Unlike the NFL or the Premier League, which historically apply strict takedown measures for copyright violations, the NBA follows a different vision. As Adam Silver summarized, "for the most part, highlights are marketing," illustrating a strategy where viral content primarily serves the league's promotion.

Faced with this volume, the NBA adopts a "laissez-faire" approach, preferring to tolerate this copyright gray area rather than risk a total loss of attention from an audience sometimes reluctant to the costs and duration of full games. This tolerance is based on a logic of audience segmentation: "Shadow Media" acts as a hook for viewers who do not subscribe to premium offers like the League Pass, but whose engagement remains vital. The quality of this content, often lower than licensed broadcasts with frantic edits focused only on successful possessions, helps protect the value of the official product while maintaining public interest.

This predominance of highlights also responds to the radical change in consumption habits, particularly within Generation Z. We observe a shift from live "television appointments" toward asynchronous and short-format consumption. Ultimately, this exposure, although not directly monetized through ticket sales or subscriptions, converts into substantial indirect revenue via merchandising and derivatives. It is precisely on these specific video streams, vectors of real but complex-to-quantify marketing value, that this project will focus

1.2. Project Objective

1.2.1. The meeting of computer vision and advertising

This project is positioned at the convergence of computer vision and advertising analytics. The central objective is to objectify and monetize the exposure generated by unofficial NBA highlights on YouTube. By leveraging state-of-the-art object detection architectures, specifically the YOLO (You Only Look Once) model, we aim to transform raw video streams into tangible economic metrics. The deployed methodology follows a rigorous sequential approach. The initial phase will consist of building and annotating a high-quality dataset to train a custom detection model, capable of identifying the complex sponsoring zones specific to NBA courts.

Once calibrated, this model will be used in the inference phase to accurately quantify not only the duration of brand appearance but also the intrinsic quality of this exposure. This visual data will then be cross-referenced with platform engagement metadata (number of views, likes, comments) to feed an econometric financial valuation formula, the foundations of which will be detailed in section 4. Finally, the project will move beyond simple retrospective analysis toward predictive modeling. By integrating a rich set of explanatory variables related to game context and player statistics, we will seek to anticipate the viral potential and future media value of upcoming matchups.

However, the very nature of this content poses a major challenge for the sustainability of the analysis: although the NBA considers these highlights as a marketing tool, their status makes them vulnerable. The sudden disappearance of certain videos or channels from the YouTube platform complicates the creation of an exhaustive history. This data volatility constitutes a significant structural challenge, which could impact the accuracy of predictive models when analyzing periods where certain video streams have been deleted.

1.2.2. Existing solution

The media audit and valuation market is dominated by reference players such as Nielsen Sports or Relo Metrics, whose expertise is now essential for major sports leagues. As illustrated by the collaboration between Relo Metrics and the Charlotte Hornets (an NBA franchise), these companies offer robust solutions to measure the return on investment of partnerships, covering both official television broadcasts and authorized digital channels. Although these players now include social media analysis in their scope, their methodologies generally prioritize content controlled by rights holders or official accounts, often leaving the granular analysis of unofficial community-generated streams on the periphery.

It is precisely on this specific segment that this project builds its unique value proposition. By focusing exclusively on unofficial highlights broadcast on YouTube. This approach allows for the quantification of massive organic exposure, thus offering franchises an unprecedented negotiation lever for their sponsorship contracts. By proving that a brand's image radiates far beyond official channels, it becomes possible to upwardly reevaluate the real value of advertising assets, while precisely identifying the areas of the court and the teams that benefit from the best quality of exposure in these short formats.

Beyond a simple retrospective measurement, this project opens the way to a much more fluid and dynamic management of advertising space. While the sponsorship economy is traditionally rigid, the contribution of a predictive model allows for the consideration of an evolving pricing system, particularly for LED panels. One could then imagine monetization based on the virality potential of a matchup: brands would

pay a premium rate for high-intensity moments likely to be captured forever, thus transforming ephemeral visibility into a lasting digital asset among younger generations.

Finally, this analysis offers a surgical understanding of placement performance. By isolating the value of each zone during key moments of the game, advertisers can optimize their presence to specifically target the asynchronous social media audience, whose consumption habits differ radically from traditional television. Ultimately, such a tool not only justifies current investments based on the real impact on digital culture.

2. Data Collection and Processing

The performance of machine learning models being intrinsically linked to the richness of their training, the creation of a robust dataset constitutes the foundation of this project. The objective is to build a database that is not only of high quality but also characterized by high variance in order to neutralize potential model biases. The methodology detailed below aims to ensure the exhaustiveness of the situations encountered, thus ensuring optimal generalization capacity during the inference phase on new games.

2.1. Origin and nature of the data

Our hybrid architecture requires the convergence of two distinct types of data to feed our models. On one hand, we collect unstructured data, in the form of high-resolution images, intended for annotation and training of the Computer Vision model for detecting sponsorship zones. On the other hand, we aggregate structured tabular data, including game history, team statistics, and individual player metrics, which will serve as explanatory variables for the predictive component of the valuation.

2.1.1. Collecting images: Video extraction pipeline

The primary source of our visual data is the YouTube platform, the exploitation of which has been automated via a Python pipeline integrating the yt-dlp library for URL extraction and video stream downloading. Image processing was then handled by the OpenCV library, allowing for the sampling of dozens of frames per video to constitute the training and testing sets. We specifically targeted the channel "TheGametimeHighlights", which is the essential reference in the field for its exhaustive coverage. This strategic choice guarantees us a substantial volume of videos covering both the current season and previous seasons, with sufficient image quality for accurate labeling.

To evaluate the model's robustness in real-world conditions, we adopted a strict temporal separation strategy: all games from the 2024/2025 season constitute the training set, while the beginning of the 2025/2026 season is reserved for the testing phase. This approach allows the model to be confronted with courts and game configurations it has never "seen" during its learning. Although the visual data shows a certain stationarity from one year to the next (courts and sponsor locations remaining relatively stable), this method validates the model's ability to generalize to familiar but technically unseen environments. This initial dataset was then subject to iterative

enrichment via secondary sources, the methodology of which is detailed in the next section.

2.1.2. NBA API

For the tabular component, we interfaced our pipeline with the official NBA API, guaranteeing access to reliable and standardized statistical data. The collection covers all games of the 2024/2025 season as well as the ongoing 2025/2026 season. This structured data is broken down into two distinct relational tables: the first aggregates game metadata (dates, matchups) and overall team statistics, while the second offers finer granularity at the player level. The latter details notably the composition of the starting five, positions, effective playing time, and complete individual performances. Relational consistency between these two databases is ensured by using the unique `GAME_ID` identifier as a join key. Finally, this raw corpus served as the basis for a feature engineering phase, allowing for the creation of complex derived variables whose relevance will be explained in the section dedicated to predictive modeling

2.2. Labeling Methodology

With a raw corpus consisting of nearly 7,000 extracted images, the next step was to transform this raw material into a high-fidelity annotated dataset. This labeling phase is critical, as it defines the "Ground Truth" on which the supervised learning of our models relies.

2.2.1. Definition of zone taxonomy

To streamline the detection process and avoid the unnecessary complexity of individually recognizing each brand, we opted for a semantic segmentation approach by zones of interest. We thus defined a taxonomy of five distinct logo classes, completed by the detection of the "ball" object. The first class, `Mid_court_logo`, isolates the central markings of the court; these are the largest assets, generally reserved for the Naming Rights partners of the venues, such as the Crypto.com Arena or the Frost Bank Center. On the periphery, we distinguish `Sid_court_logo`, modest-sized static markings located near the benches and absent in some franchises, from `Sid_court_led_logo`. The latter, corresponding to side LED panels, are of major strategic importance: their dynamic nature allows for the rotating display of multiple

advertisers and their positioning offers optimal visibility during transition or steal phases.

The baseline area is covered by the Back_court_logo class. This category presents a specific challenge for the model, which must learn to discriminate commercial sponsors from institutional markings (team name), omnipresent on the baseline but excluded from our valuation. Finally, we integrated the basket_logo class to capture the multitude of small logos placed on the basket structure (stanchion). Although small in size, these locations benefit from critical overexposure during slow-motion action at the rim. In addition to these commercial zones, a sixth basketball class was added to track the ball, whose position will later serve as a proxy to model the viewer's visual point of attention.



Figure 1: Screenshot of Label-Studio used for manual data labeling. The image notably shows an example featuring the various classes.

2.2.2. Human-in-the-Loop Labeling

The inherent complexity of the task, which required the precise annotation of about ten oriented boxes per image, necessitated an optimization strategy to build a large dataset within reasonable timeframes. We therefore implemented an iterative "Human-in-the-Loop" approach. The process began with a bootstrapping phase consisting of the entirely manual annotation of approximately one hundred images using the Label Studio tool. This initial core allowed for the training of a first embryonic model, which was immediately deployed to pre-annotate subsequent batches of images. This transition toward correcting predictions, rather than creating them from scratch, considerably accelerated the workflow while reducing the annotator's cognitive load.

The advantage of this method goes beyond simple time productivity gains. Pre-annotation acts as a safety net, minimizing inattention errors (human false negatives) often induced by the repetitive nature of the task. Furthermore, this iterative cycle allowed human effort to be focused on edge cases and model failures. By actively correcting "bad" predictions rather than simply validating the obvious ones, we were able to identify the model's contextual biases early and guide the data enrichment strategy. This process was repeated until a robust training corpus of nearly 2,000 annotated images was obtained

2.2.3. Iterative Data Enrichment

As soon as the primary source reached a saturation point in terms of visual variance (around 500 to 600 images), we initiated a qualitative enrichment phase to diversify exposure scenarios. We first integrated official highlights from the NBA YouTube channel covering the last three seasons (2022-2025). These sequences, often longer and more cinematographic, allowed the model to be exposed to a greater variety of camera angles (zooms, wide shots, slow motion) and to all the league's courts, thus breaking the monotony of standard fixed shots.

In a second stage, the effort focused on reducing false positives (hallucinations) by confronting the model with environments having a low density of logos. We integrated G-League games, where the courts are significantly less commercially saturated, to teach the model that the presence of sponsors (notably on the basket or at mid-court) is not systematic. This "hard negatives" logic was pushed further with the addition of images from amateur courts (indoor and outdoor) devoid of any markings, as well as out-of-domain images (cities, NFL/NHL stadiums). It is crucial to note that these exogenous data were strictly confined to the training and validation sets; the test set, for its part, remains faithful to the actual distribution of target data to guarantee an integral evaluation of production performance

2.3. Exploratory Data Analysis

The development of the dataset followed a rigorous iterative process, totaling ten distinct versions. This evolution is split into two major phases: an initial sequence (versions 1 to 6) focused solely on sponsoring zones, followed by a second (versions 7 to 10) integrating the "ball" class. This structural distinction explains the absence of ball-related data in the initial comparative graphs. To illustrate the maturation of the corpus, we propose here a comparative analysis between the embryonic version (v1, approximately 200 images) and the final version (v10, approximately 2,000 images).

The analysis of the area distribution (in pixels) validates the physical consistency of our taxonomy. As anticipated, the `mid_court_logo`, `back_court_logo`, and `side_court_led_logo` classes present the largest areas, reflecting their actual predominance on the court. Conversely, `basket_logo` and `side_court_logo` occupy a marginal place on the screen. The fundamental contribution of successive iterations can be seen in the evolution of the standard deviation: the final version shows a significantly more spread-out distribution for the majority of classes. This increased variance testifies to the diversity of contexts and camera angles integrated (zooms, wide shots), confirming that the model is now exposed to a wider range of viewpoints. This observation is corroborated by the analysis of the bounding boxes' orientation, whose broader distribution in the final version indicates better coverage of camera angles.

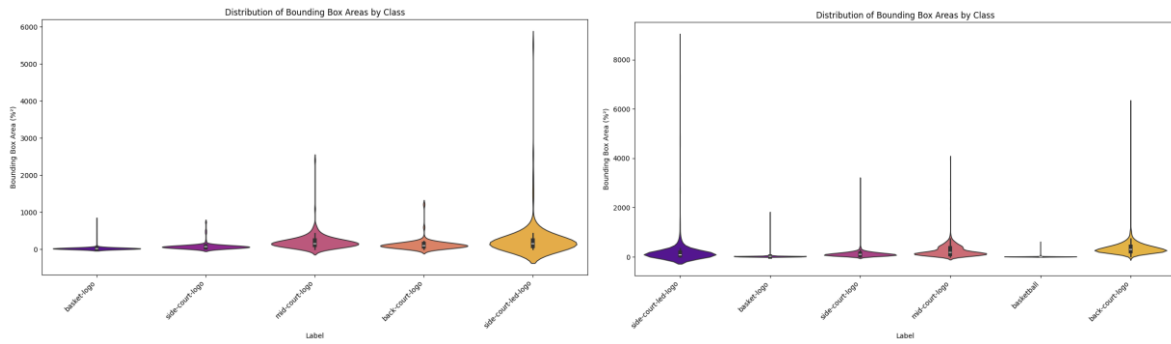


Figure 2: Distribution of the sizes of the different zones in the first version of the database (left) versus the final version (right).

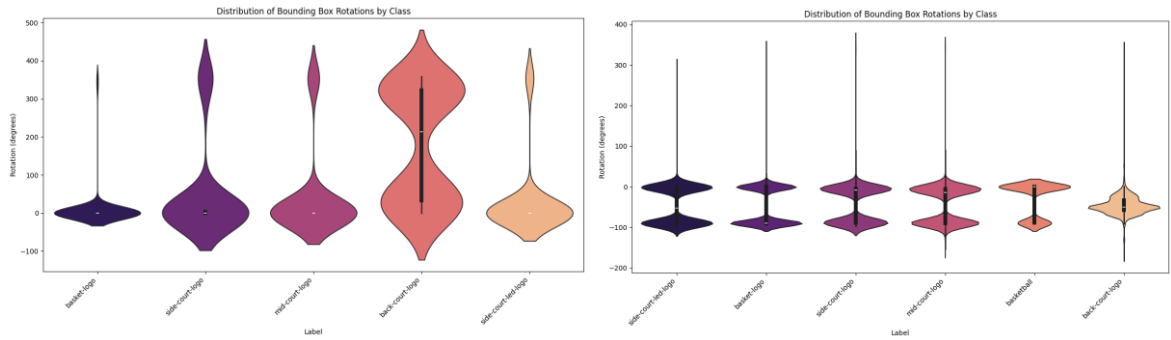


Figure 3: Distribution of rotation for the various zones in the first version of the database (left) versus the final version (right).

Managing class balance was a constant challenge during annotation. The `basket_logo` class suffers from an inevitable structural overrepresentation because the basket is the focal point of the game and often features multiple simultaneous brandings. While manual filtering of these occurrences became prohibitively time-consuming as volume increased, tests show that this numerical preponderance did not negatively impact the model's convergence on other classes. not negatively impact the model's convergence on other classes.

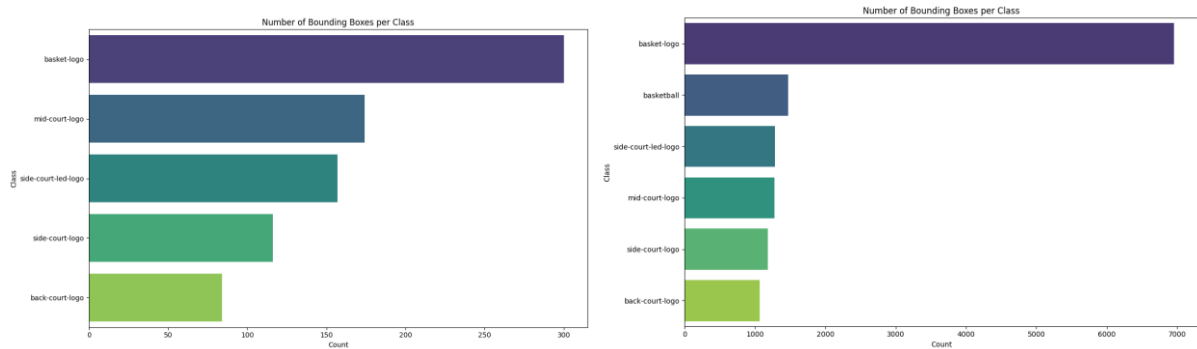


Figure 4: Distribution of classes in the first version of the database (left) versus the final version (right).

Spatially, the position heatmap illustrates the completeness of the coverage achieved in version 10. Unlike early iterations, almost the entire visual space is covered, with the notable exception of a lower horizontal band. This gap is not a collection bias but a structural constraint of sports broadcasting: this area is systematically obscured by the score overlay, making the presence of logos in this specific location difficult as they are hidden

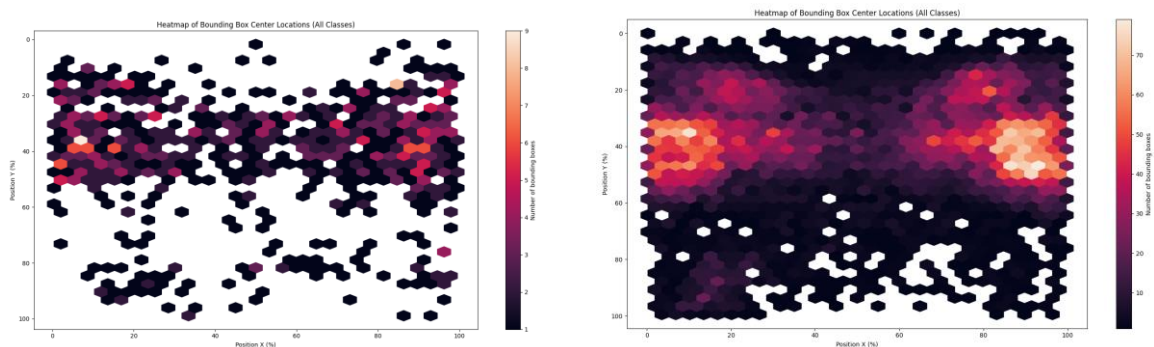


Figure 5: Distribution of detection positions in the first version of the database (left) versus the final version (right).

To conclude this exploratory analysis, spatial density maps were generated for each class in the final version of the dataset (v10). These visualizations confirm the geometric consistency of the annotations relative to the structure of a basketball court. The classes exhibit very distinct spatial signatures: the `side-court-led-logo` and `side-court-logo` classes form distinct horizontal bands, reflecting their placement along the sidelines. Simultaneously, the `basket-logo` and `back-court-logo` classes show a clear

bimodal distribution with two lateral concentration poles, corresponding to the two baskets located at the ends of the court as captured by standard broadcast cameras. The mid-court-logo class, for its part, presents more dispersed but localized clusters, consistent with different camera angles of the center circle.

The most critical observation for the subsequent modeling concerns the basketball class. Unlike the fixed sponsorship zones, the ball's heatmap reveals a quasi-Gaussian distribution, forming a dense cloud concentrated around the center of the image and fading radially. This empirical validation supports the "centrality of attention" hypothesis: on average, the action takes place in the center of the screen. This finding justifies the methodological choice applied during the quantification of exposure quality: in the event of losing the ball's tracking for several consecutive frames, the center of the screen will be used as a reliable proxy to estimate the spectator's point of gaze fixation and calculate the relative distance of the logos.

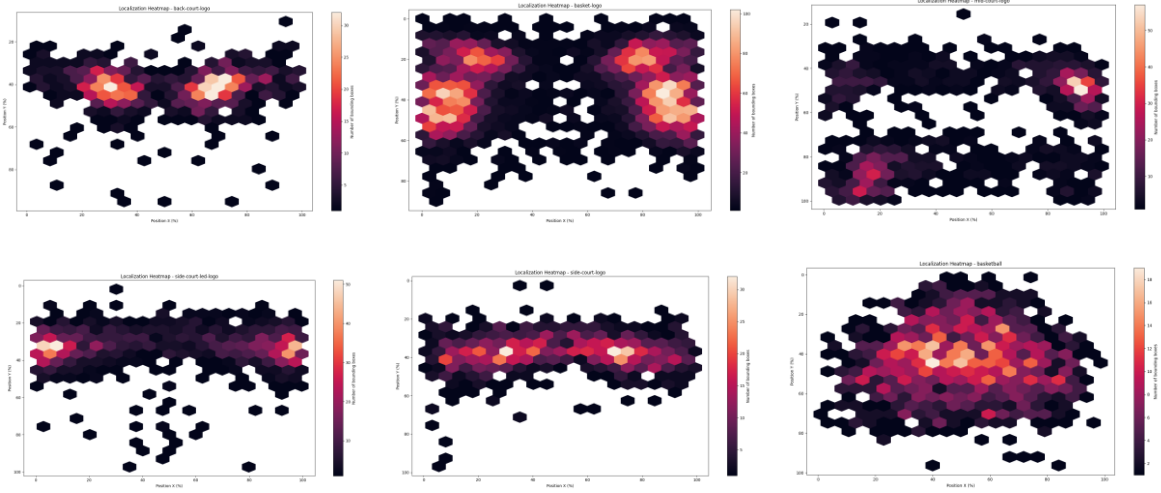


Figure 6: Distribution of detection positions for each of the 6 classes for the final version of the database.

3. Visual Detection Architecture

The quality of economic predictions depends intrinsically on the accuracy of the extracted data, making the choice of detection architecture a critical step. This section details the selected model, justifies its suitability for the specific constraints of sports broadcasts (camera angles, deformations), and explains the training strategy adopted to maximize performance on our iterative dataset.

3.1. Model Selection

To address the issue of sponsor quantification, an Object Tracking approach would have been the most methodologically sound path. Indeed, tracking ensures native temporal continuity, allowing the unique identity of a logo to be maintained across frames and managing momentary occlusions more finely. However, this approach introduces increased computational complexity. For the sake of pragmatism and simplicity in this first iteration, we prioritized a frame-by-frame Object Detection strategy. Although simplified, this method offers immediate robustness and sufficient execution speed to estimate cumulative exposure volumes, accepting the compromise of treating each image as independent data.

The technological choice fell on the current state-of-the-art: the YOLO architecture in its version 11, specifically its OBB (Oriented Bounding Box) variant. This selection is dictated by the particular geometry of sports markings and camera perspectives. Unlike classic axis-aligned bounding boxes, which are parallel to the edges of the image, oriented boxes introduce an angle parameter. This feature is essential for cleanly outlining logos located in the backcourt or filmed from a high angle. Where a classic box would capture up to 50% parasitic pixels around a tilted logo, the OBB fits the exact shape of the marking.

Finally, operational scaling constraints validated this architectural choice. Since our pipeline must process thousands of ten-minute videos, the performance-to-speed ratio is a critical factor. YOLOv11 currently offers the best compromise between mean average precision (mAP) and inference latency per millisecond, making large-volume processing economically viable. Furthermore, given the volume of our annotated dataset (approximately 2,000 images), using the lightweight versions of the model (Nano, Small, Medium) proved to be the most relevant to prevent overfitting while ensuring optimal execution speed.

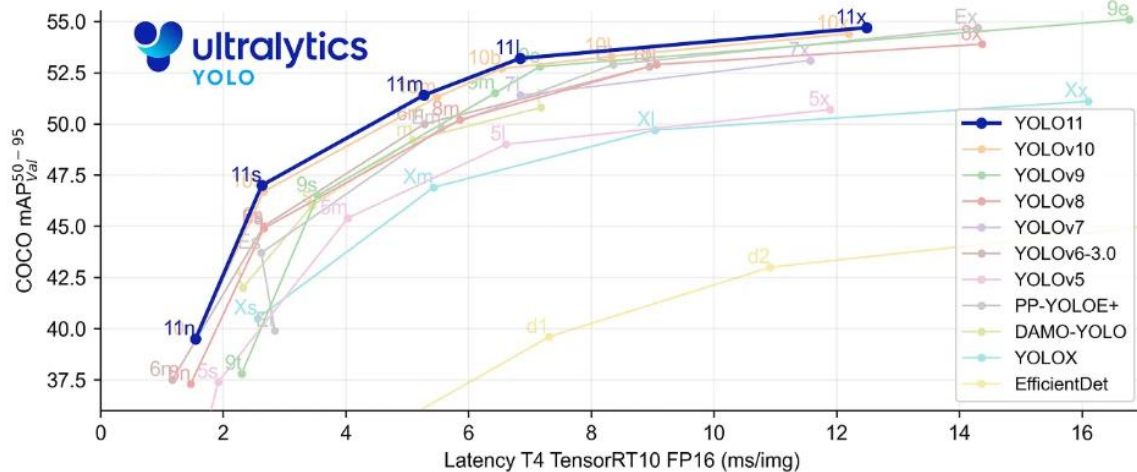


Figure 7: Comparison of the speed-performance trade-off for the main object detection models. Source: Ultralytics.

The uniqueness of the YOLO algorithm lies in its "one-stage" architecture that treats object detection as a single mathematical regression problem, whereas traditional methods separate localization and classification. Specifically, the model divides the image into a grid and simultaneously predicts for each cell the presence of an object and its geometric properties. In our specific OBB configuration, the network calculates in a single pass the center coordinates, dimensions, confidence score, and especially the object's rotation angle, allowing it to perfectly fit the shape of tilted logos on the court.

3.2. Training Strategy and Fine-Tuning

The training protocol relies on rigorous data segmentation into training (train) and validation (val) sets, ensuring the integrity of performance evaluation. A comparative study of model sizes was conducted to identify the architecture offering the best performance-to-computational-cost ratio. Initial tests demonstrated that the YOLOv11n (Nano) model, although very fast, lacked generalization capacity for small objects. Conversely, switching to the YOLOv11s (Small) model resulted in a significant gain in precision (mAP), amply justifying the slight increase in inference time. Experiments with heavier architectures (Medium/Large) showed only marginal gains that did not compensate for the drastic increase in calculation times. The Small model therefore emerged as the optimal balance point.

Modèle	Taille (pixels)	mAP ^{test} ₅₀	Vitesse CPU ONNX (ms)	Vitesse T4 TensorRT10 (ms)	paramètres (M)	FLOPs (B)
YOLO11n-obb	1024	78.4	117,6 ± 0,8	4,4 ± 0,0	2.7	16.8
YOLO11s-obb	1024	79.5	219,4 ± 4,0	5,1 ± 0,0	9.7	57.1
YOLO11m-obb	1024	80.9	562,8 ± 2,9	10,1 ± 0,4	20.9	182.8
YOLO11l-obb	1024	81.0	712,5 ± 5,0	13,5 ± 0,6	26.1	231.2
YOLO11x-obb	1024	81.3	1408,6 ± 7,7	28,6 ± 1,0	58.8	519.1

Figure 8: Comparison of the speed-performance trade-off for the latest version of YOLO-OB. Source: Ultralytics.

A second axis of optimization focused on input image resolution (Input Size). The standard resolution of 640 pixels, used in early iterations, proved insufficient for fine detection, especially after integrating the "ball" class, which has a reduced visual signature. We progressively increased the resolution to 1024, then to 1280 pixels. This sizing allows for the preservation of high-frequency details necessary for detecting small distant logos and the ball.

Regarding the data augmentation strategy, we chose not to introduce external preprocessing, relying exclusively on YOLO's robust native pipeline. This includes Mosaic augmentation, a technique that combines four training images into one according to random ratios. This forces the model to learn about objects at varied scales and in truncated contexts, significantly improving its robustness.



Figure 9: 8 examples (2 batches) from the training set featuring Mosaic augmentation.

Finally, hardware constraints dictated certain execution parameters. Training was performed on a GPU with 8 GB of VRAM. This memory limitation, coupled with the

high input resolution (1280px), required a reduction in batch size to 4. Training was conducted for 50 epochs, a duration sufficient to observe a convergence of validation metrics and the attainment of a performance plateau, confirming the effectiveness of transfer learning from pre-trained weights.

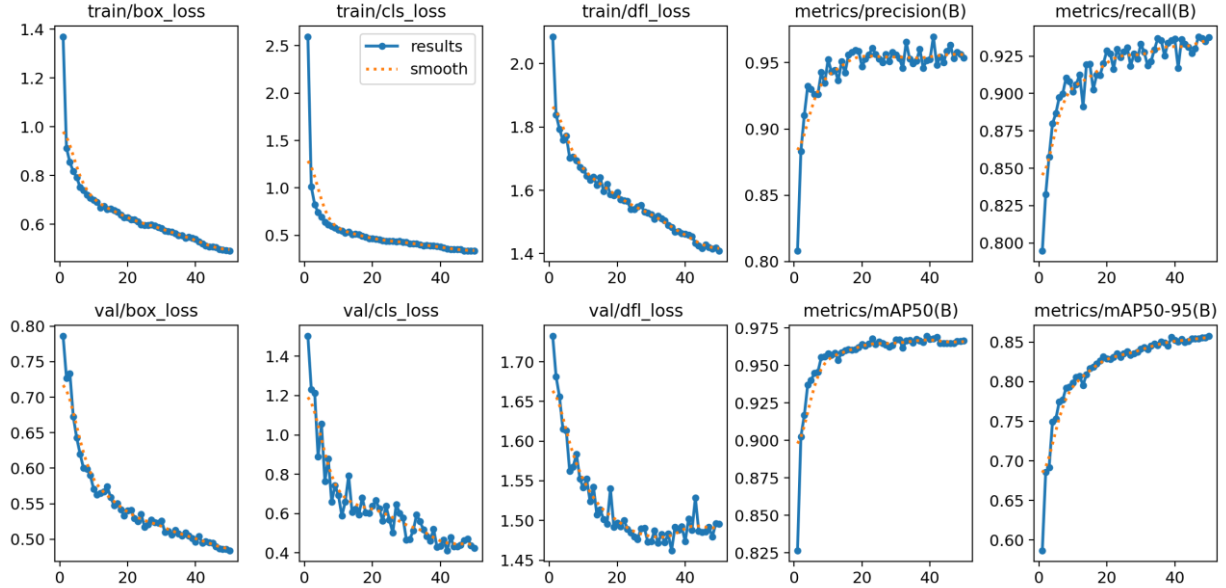


Figure 10: Learning curves of the model's final iteration.

3.3. Performance Evaluation and Error Analysis

3.3.1. Metrics Analysis

The evaluation of our model is not limited to raw detection maximization but responds to a strict business constraint: financial credibility. From a media valuation perspective, minimizing False Positives (hallucinating a non-existent logo) is imperative. It is economically preferable to slightly underestimate exposure (False Negatives) rather than overestimate it, which would erode the trust of sponsors and the NBA. To align the model with this conservative strategy, we raised the confidence threshold required for detection to 0.6, compared to the default 0.5. This deliberate choice, visible on the Precision-Confidence curves, stabilizes precision at a high level while accepting a slight decrease in recall in return. We thus prioritize the certainty of the produced data over its absolute quantity.

The overall performance of the model is validated by the mAP50 metric (mean Average Precision at an IoU threshold of 0.5), which reaches a remarkable score of 0.950 across all classes. Fundamental in computer vision, IoU (Intersection over Union) measures the overlap ratio between the predicted bounding box and the ground truth to quantify geometric precision. Thus, mAP50 corresponds to the average of precisions obtained when considering any prediction with an overlap with reality greater than 50%

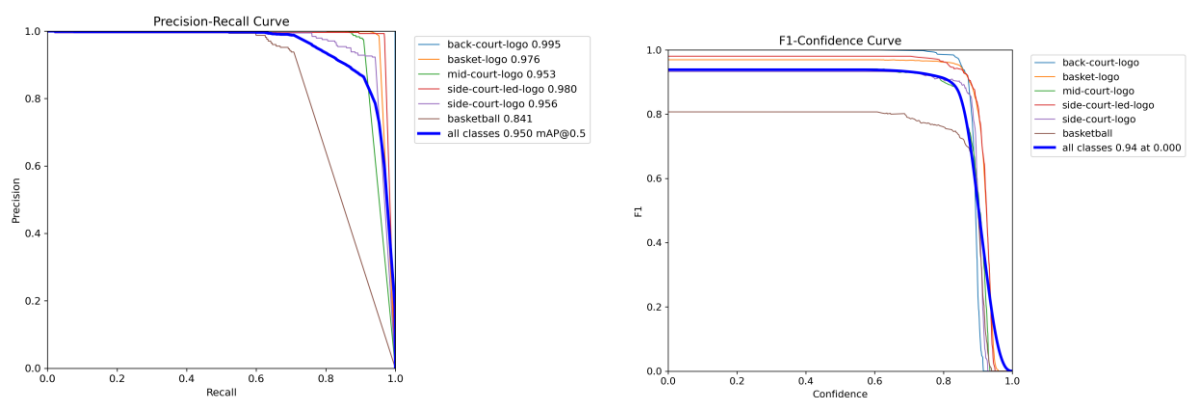
as valid. More demanding, the mAP50-95 metric stands at 0.867. Calculated by averaging mAP scores over ten progressive IoU thresholds (from 0.5 to 0.95), it penalizes clipping approximations more severely and demonstrates the great finesse of object localization here.

Detailed analysis by class reveals an expected dichotomy between static and dynamic elements. Fixed logos display near-perfect performance: the back-court-logo reaches an mAP50 of 0.995 and the side-court-led-logo 0.980. Conversely, the basketball class, subject to rapid movements and motion blur, logically presents the lowest results, although very satisfactory, with an mAP50 of 0.841

Metric	All	Back-Court-Logo	Basket-Logo	Mid-Court-Logo	Side-Court-Led-Logo	Side-Court-Logo	Basketball
mAP50	0.950	0.995	0.976	0.953	0.980	0.956	0.841
mAP50-95	0.867	0.907	0.933	0.847	0.915	0.864	0.738

Figure 11: mAP50 and mAP50-95 performance across the different classes.

The study of Precision-Recall curves confirms the robustness of the YOLOv11-OBB architecture on our dataset. The global curve maintains precision close to 1.0 over a large part of the recall axis, only dropping late. This characteristic indicates that the model manages to recover the majority of relevant objects without degrading prediction quality. Class-specific curves show that mid-court and side-court logos maintain strong structural integrity, while the basketball class curve decreases more rapidly, illustrating the inherent difficulty in maintaining high precision when trying to detect the most difficult occurrences (partially obscured or distant ball).



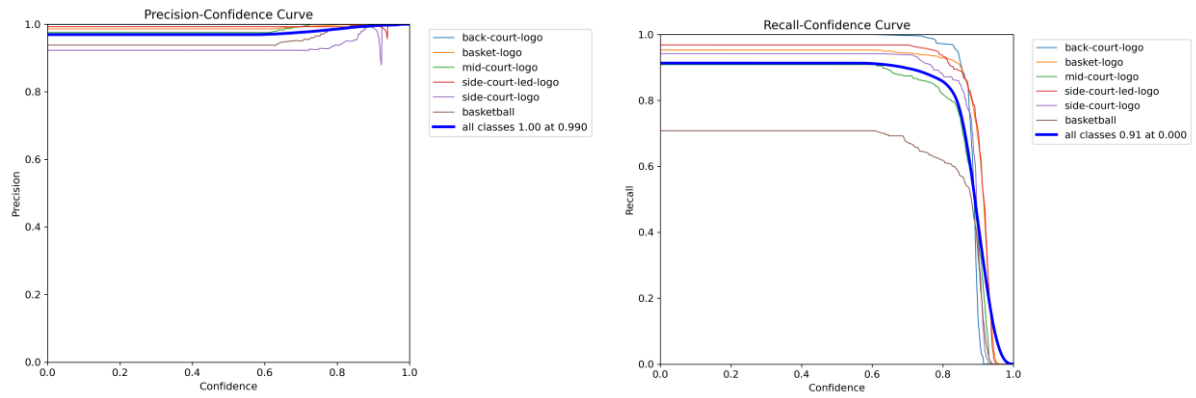


Figure 12: Precision-recall curve (top left), F1-confidence (top right), Precision-confidence (bottom left), and recall-confidence (bottom right).

Finally, the analysis of Confusion Matrices offers granular insight into model behavior. On the normalized matrix, we observe a dominant diagonal reflecting an extremely high correct classification rate for sponsorship zones: 100% for back-court-logos and 97% for side-court-led-logos. Inter-class errors are marginal, validating the relevance of our visual taxonomy. The main point of attention remains the confusion between the basketball class and the background. The matrix reveals that while the model excels at not hallucinating logos (very low False Positive rate on sponsors), it tends to "miss" the ball in certain complex configurations (False Negatives), classifying it as background. This behavior is consistent with our thresholding strategy at 0.6: we prefer to lose a few frames of ball tracking rather than generate noise in the valuation data.

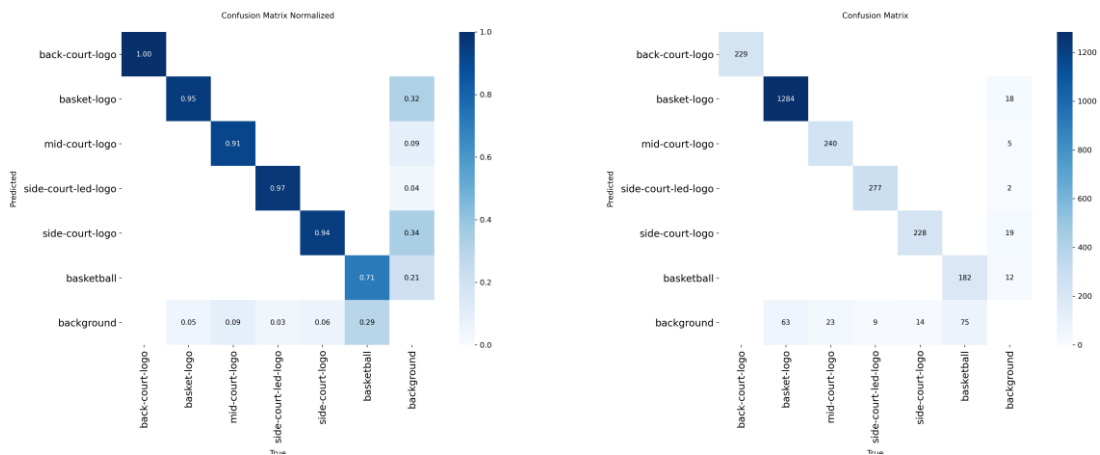


Figure 13: Raw and normalized confusion matrices of the model.

3.3.2. Error Analysis

The analysis of the confusion matrices confirms the reliability of the model on fixed elements. The back-court, mid-court, and side-court-led classes generate almost no hallucinations. Errors are mainly concentrated in the basketball class. Despite the high

confidence threshold, the model sometimes confuses the ball with round objects such as heads, knees, or elbows. This phenomenon is accentuated by the rapid movement of players, which makes the image blurry. We also notice a significant number of missed logos in the center circle (mid-court-logos). This is often due to the score overlay, which hides part of the logo and prevents the model from validating it with confidence.

However, these imperfections can be corrected using simple logical rules applied during post-processing. For the ball, we start from an obvious fact: there is only one ball in play. If the model detects several balls in the same image, we only keep the one with the highest probability score. This rule automatically eliminates the majority of false detections that could distort the attention analysis.

Finally, we correct the problem of logo fragmentation. During fast camera movements (panning), the model tends to cut a long LED panel into several distinct boxes. Since two logos of the same class cannot physically overlap, we systematically merge boxes of the same class that overlap. These corrections, applied after inference, allow us to "clean" the data and ensure its consistency before feeding it into the econometric model.

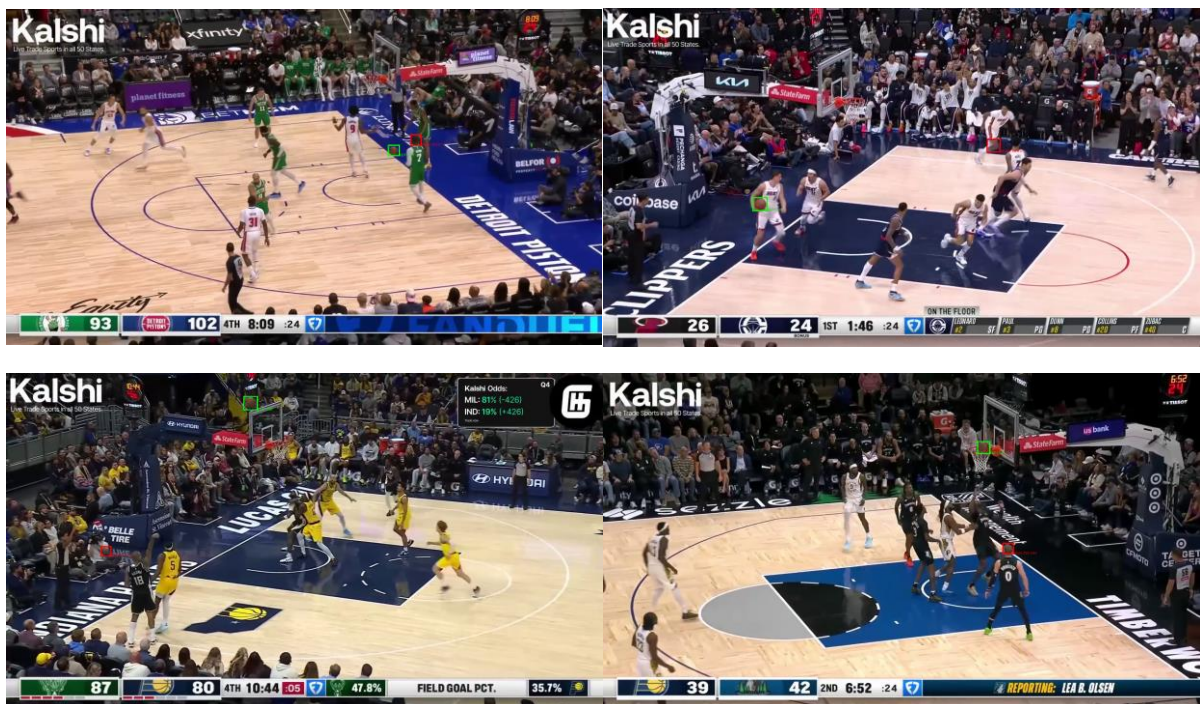


Figure 14: Some examples of corrected ball detection. In green is the retained detection, the one with the highest probability of being the ball, and in red is the deleted prediction.

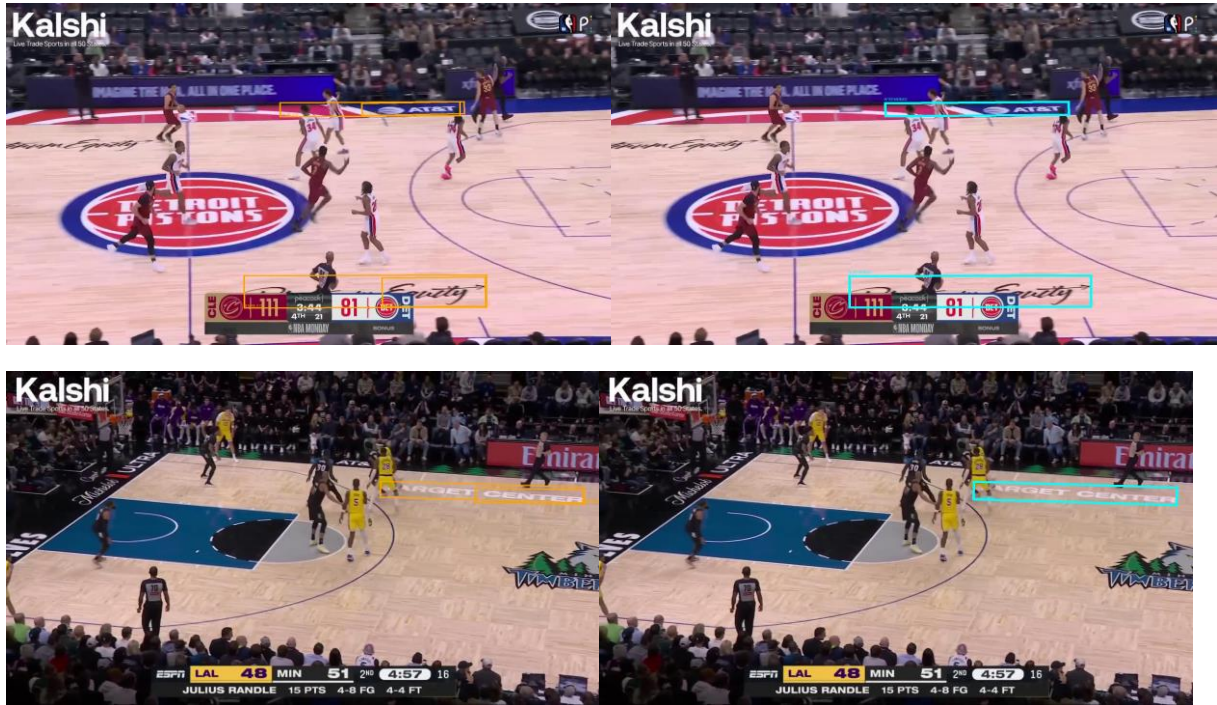


Figure 15: Two examples before and after the merging of detections during overlaps.

3.3.3. Interpretability

To overcome the opacity inherent in deep neural networks, we integrated an interpretability layer using the EigenCAM algorithm. Unlike classic gradient-based methods (such as Grad-CAM), EigenCAM relies on the calculation of principal components (PCA) of the spatial activation maps generated by the last convolutional layer of the model. Specifically, this method projects the high-dimensional features learned by the network onto a 2D heatmap, thus revealing the areas of the image that contribute most to the detection decision. This allows us to visualize precisely what the model is "looking at" to justify the presence of a sponsor.

The analysis of the generated heatmaps reveals a fascinating and counter-intuitive behavior: activation is not restricted exclusively to the textual morphology of the logos. As observed in the visualizations, the heat zones largely encompass the immediate contextual environment: boundary lines (3-point line, paint), the physical structure of the basket, and even the first rows of the stands or the players' bench. The model does not merely read a logo; it analyzes the global scene.

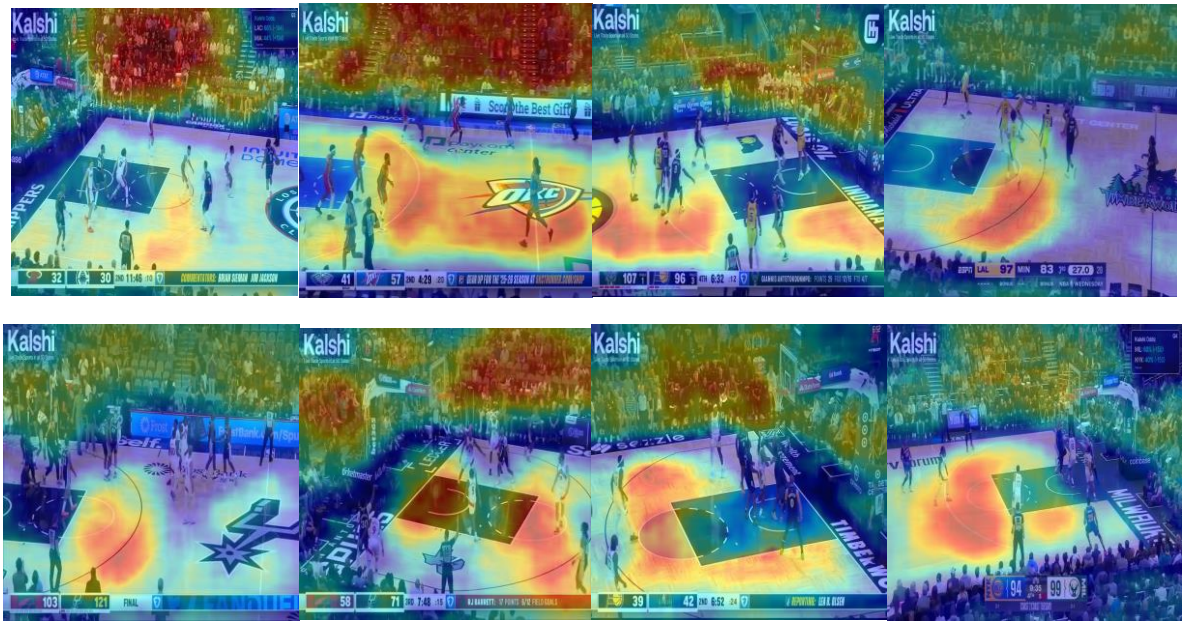


Figure 16: Some examples from the test set of EigenCam.

4. Econometric Modeling of Media Value

The initial phase of this project developed a robust computer vision architecture capable of segmenting key advertising spaces on an NBA court. This fourth section focuses on the operational use of this tool to objectify the economic value generated by unofficial highlights. The analysis corpus is built using an automated extraction pipeline. A dedicated script performs a YouTube search based on official NBA data (dates, matchups) to identify relevant content. Result filtering applies strict criteria to the top ten results: temporal consistency, exclusion of official rights holders' channels, a minimum engagement threshold (1,000 views), and duration constraints. The detailed methodology for this collection is provided in the appendix.

4.1. Use of the model

Inference is performed on the collected video streams primarily using the YOLOv11-Small model, identified during testing as the best compromise between detection accuracy and execution speed. To optimize computational load, we apply temporal downsampling by processing the stream at 5 frames per second (FPS) instead of the native 30 or 60 FPS. This heuristic approximation, statistically justified in the appendix, drastically reduces calculation times without degrading the representativeness of the measurement. In real conditions, processing an average 9-minute sequence takes approximately 30 seconds. However, scaling this to a full regular season of 1,230 games, each generating more than three exploitable sources, represents a computational cost that must be managed.

4.1.1. Quantity of exposure

The fundamental metric of our valuation model is the gross duration of exposure per zone. It serves as the quantitative pillar for the financial evaluation. The calculation stems directly from the 5 FPS sampling frequency: each processed frame represents a temporal increment of 0.2 seconds. Cumulative exposure for a given zone is obtained by dividing the total number of detections by the sampling frequency.

4.1.2. Quality of exposure

While the raw measurement of exposure time constitutes the bedrock of our analysis, it remains an incomplete metric without qualitative weighting: a second of blurry

visibility on the periphery does not have the same economic value as a clear second at the heart of the action. To objectify this quality, we extract a vector of visual characteristics frame by frame, the primary one being the centrality of attention. We use the position of the ball as a proxy for the spectator's gaze fixation point: the spatially closer a logo is to the ball (Euclidean distance), the stronger its supposed cognitive impact is. However, as ball detection is subject to interruptions (occlusions, kinetic blur), we have implemented a temporal continuity heuristic to robustify this metric. When no ball is detected, we impute its position using its last known coordinate over a sliding window of 1.5 seconds, based on the principle that attention persists briefly on the last zone of action. Beyond this tolerance threshold, if the ball remains invisible, the model switches by default to the geometric center of the screen as the attentional reference point.

In addition to this spatial dynamic, we evaluate three other physical indicators that are decisive for advertising impact. First, the surface area of the bounding box allows us to quantify the spatial occupation of the logo: intuitively, a larger surface area maximizes the probability of perception. Second, we calculate the Laplacian Variance of the region of interest to measure image sharpness (signal integrity). This step is crucial for filtering out unreadable occurrences which, although detected, provide no brand memorization value. Finally, we integrate a measure of competitive density, or local "Share of Voice." This indicator evaluates the visual clutter around the target logo: a sponsor's impact is mechanically diluted if it appears in a dense cluster of other brands, as the spectator's attention is then fragmented.

4.2. Financial valuation formula

Having defined and extracted these raw metrics (distance, size, sharpness, clutter), the next step is to aggregate them into a coherent economic model. This section justifies the mathematical choices made to transform these physical indicators into a normalized "Quality Score" and details how this score articulates with audience volumes to produce a final monetary valuation.

4.2.1. Simple formula

We first present the formula actually deployed in this project. It is a intentionally simplified approach compared to the multitude of exogenous variables an exhaustive model could theoretically integrate. This formulation has structural limits and inherent biases that will be addressed when discussing a more "ideal" formula later.

4.2.1.1. *Cost per View (CPV) Calibration: Market Benchmarking*

The methodology used to convert raw data (such as view counts and exposure time) into monetary value consists of establishing an equivalence based on the Cost per View (CPV) per second of a dedicated advertising space. This reference value is volatile, fluctuating by platform and season timing. Relative prices naturally increase during the playoffs due to massive engagement.

For example, on a major channel like ESPN, determining the reference CPV per second involves dividing the cost of a commercial by the average audience and the spot's duration. If a 30-second spot costs \$150,000 for an audience of 3 million, the CPV is €0.05. This results in a CPV per second of \$0.0016.

For this YouTube-focused project, we use a similar reasoning based specifically on the cost of an integrated product placement. We deliberately excluded standard pre-roll ad rates in favor of commercial collaborations with content creators. This approach better reflects brand integration into content, which typically commands higher valuation due to audience trust in the creator. Market analysis led us to a reference value of \$0.00033 per view per second of exposure as the base of our formula.

This figure is determined by the specific sports niche where engagement is higher than in "lifestyle" sectors, though lower than in financial themes. CPM benchmarks indicate this value also depends on channel size and global engagement (watch time). Finally, while the video sector defines the viewer's value, the brand's sector is also crucial; industries like finance and sports betting, heavily present in the NBA, are willing to invest significantly more to capture this audience.

4.2.1.2. Quality Index: Modeling Visual Attention

It is economically inconsistent to bill incidental exposure at the same rate as a dedicated commercial, as the quality of attention is significantly different. To weight this raw value, we calculate a normalized score between 0 and 1 called the "Quality Index". The purpose of this index is to mathematically penalize the exposure of a logo that has a low probability of being perceived or remembered (brand recall). This composite index results from the product of four distinct sub-scores detailed below.

4.2.1.2.1. Visual Saliency: Sigmoid Transfer Function

The first determinant of the quality index is visual saliency, objectified by the relative area of the detected bounding box (expressed as a percentage of the screen). The relationship between a logo's physical size and its cognitive impact is non-linear: below a critical resolution threshold, readability is zero and retention is impossible;

conversely, beyond a certain size, the marginal gain in attention becomes negligible (law of diminishing returns). To model this threshold dynamic and progressive saturation, a sigmoid transfer function is used:

$$f(x) = \frac{1}{1 + e^{-c_1 \cdot (x - c_2)}}$$

The modeling challenge therefore lies in the fine calibration of the curve's hyperparameters: the inflection point (centering) and the steepness of the slope. This calibration stems directly from the exploratory analysis of the area distributions within our dataset. Empirical observation reveals a high heterogeneity of median areas according to the classes: approximately 0.2% for logos on the basket (basket-logo), compared to 0.8%, 1.2%, and 1.5% respectively for the side-court, side-court-led, and back-court, up to 1.8% for the center circle.

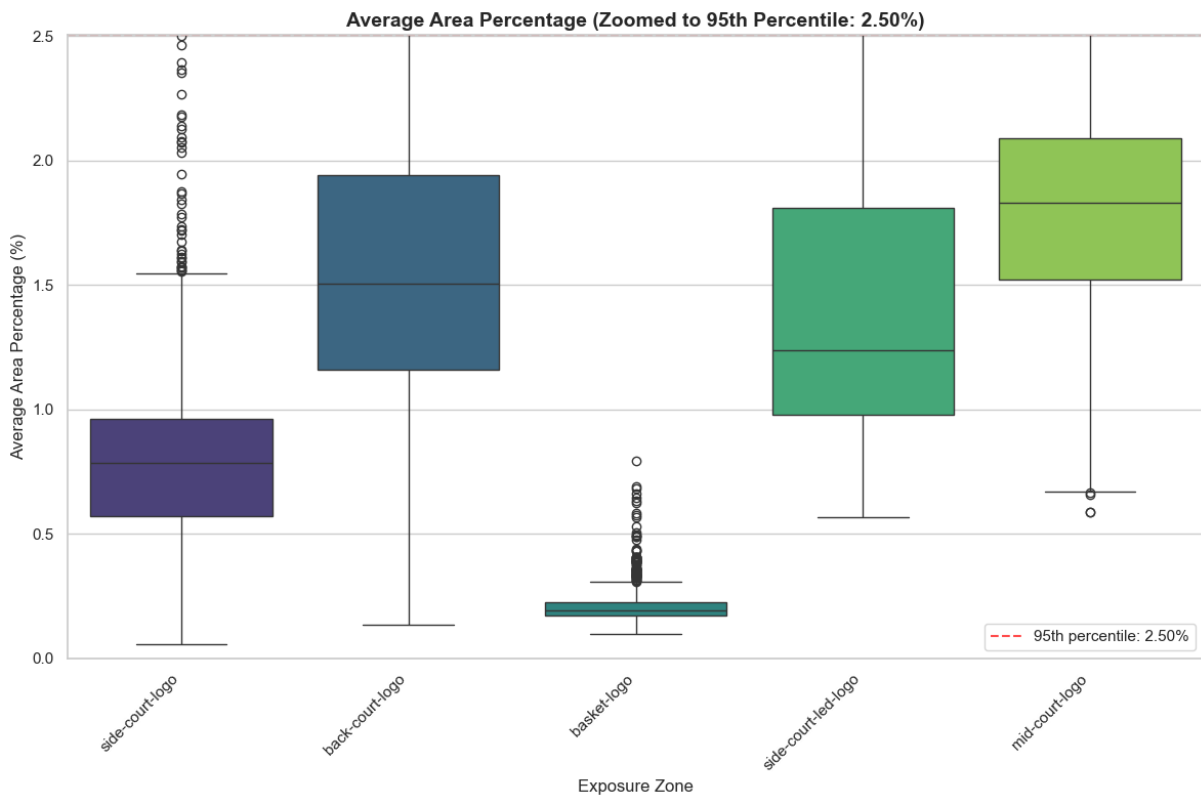


Figure 17: Size distribution of logos detected by the model for each zone of the court.

he parameterization strategy aims to not excessively penalize basket logos which, despite their small surface area (0.2%), benefit from excellent readability due to their contrast and focal positioning. We have therefore fixed the pivot point (central value where the score reaches 0.5) at a surface area of $c_2 = 1.5\%$, corresponding to the standards of major floor markings. To control saturation, we opted for a gentle slope defined by a coefficient $c_1 = 100$. This choice, although empirical, is justified by the need to smooth the value curve: it allows for a significant score to be assigned to small

readable formats, while avoiding the premature granting of a maximum score (1.0) to logos barely exceeding 2% of the screen. A logo covering 3% of the surface, although very visible, cannot indeed be equivalent in terms of absolute impact to a full-screen advertisement.

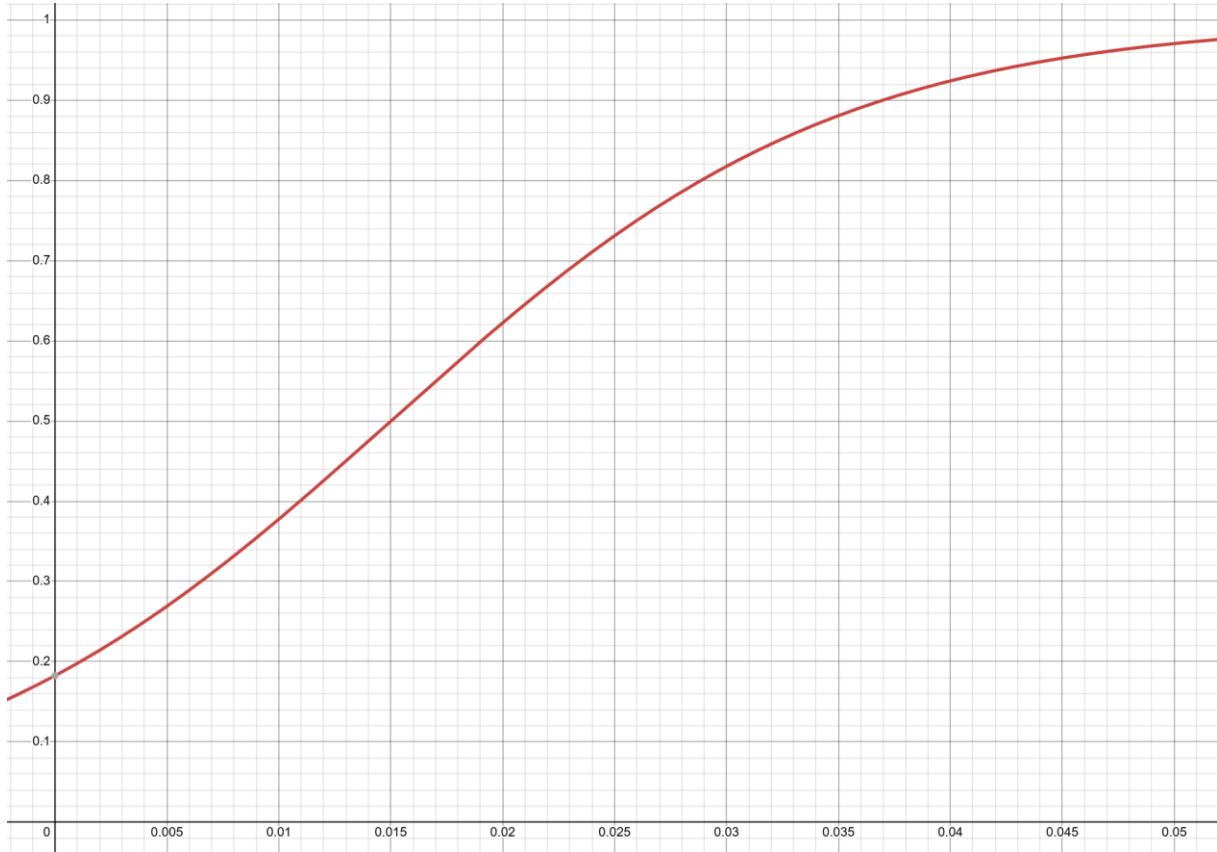


Figure 18: Illustration of the function used. The y-axis represents the score between 0 and 1, and the x-axis represents the size as a percentage of the screen.

4.2.1.2.2. Spatial Centrality: Gaussian Decay

The second pillar of valuation is spatial centrality, calculated from the normalized Euclidean distance between the geometric center of the logo and that of the ball (expressed as a percentage of the screen's diagonal). To model the decay of attention as one moves away from the focal point, we use a Gaussian function centered on the ball. In this model, the standard deviation sigma acts as the control parameter for the "attentional spotlight," defining the width of the active perception field

$$f(x) = e^{-\frac{x^2}{2\sigma^2}}$$

If a logo is located at the heart of the distribution, its probability of perception and memorization is maximal. Conversely, on the periphery of this attentional focus, the value decreases exponentially. Academic literature in cognitive science generally

suggests narrow focus values (around $\sigma = 0.10$) for pure object-tracking tasks. However, we have opted for an empirical approach adapted to the dynamics of basketball, setting $\sigma = 0.20$.

This deliberate choice to broaden the attentional spectrum is justified by the very nature of sports broadcasting. Unlike a static laboratory task, the "average" spectator's gaze (non-expert tactician) does not remain locked on the ball. Attention navigates dynamically within a wider action zone, encompassing the body movements of the ball handler, the positioning of their direct defender, and the movements of nearby teammates. By adopting a σ of 0.20, we integrate this peripheral tolerance, recognizing that a logo located in the immediate vicinity of the game action benefits from valuable exposure, even if it is not strictly aligned with the ball.



Figure 19: Illustration of the function used. The y-axis represents the score between 0 and 1, and the x-axis represents the distance to the ball as a percentage of the screen diagonal.

4.2.1.2.3. Visual Clutter: Weighted Share of Voice

Another determining measure of quality is visual clutter. Unlike a classic commercial where the brand enjoys total exclusivity on the screen, the reality of sports sponsorship imposes space sharing. Spectator attention inevitably dilutes when our logo is drowned in a dense area, surrounded by a multitude of competing sponsors. We seek here to quantify this phenomenon via a local "Share of Voice" formula. The objective is to

calculate the relative importance of the target logo compared to its immediate environment, to penalize situations of visual saturation.

A naive approach would consist of simply dividing the area of the target logo by the sum of the areas of all logos present on the screen. However, this method has a major bias: it attributes the same harmful weight to a competing logo stuck to ours as to a very distant logo, potentially invisible or outside the field of attention. To correct this, we weight the size of competing logos by their distance to the logo of interest. We reuse the Gaussian decay function:

$$f(x) = \sqrt{\frac{Area_{cible}}{Area_{cible} + \sum_{i=1}^n \left(Area_i \cdot e^{-\frac{x_i^2}{2(0.10)^2}} \right)}}$$

but with a stricter dispersion parameter (sigma=0.10). This technical choice allows us to consider as truly "distracting" only those logos located in the immediate vicinity. Finally, to prevent this function from being too punitive and generating excessively low scores at the slightest competition, we apply a square root to the overall result. This allows for a mechanical boost of the scores and a smoothing of the penalty curve, as illustrated by the example below simulating a logo confronted with a competing logo measuring 3 times the size of the target logo. The formula is therefore as follows:

$$f(x) = \frac{0.01}{0.01 + 0.03 \cdot e^{-\frac{x^2}{2(0.1)^2}}}$$

The x-axis here represents the distance of the competing logo from the target logo.

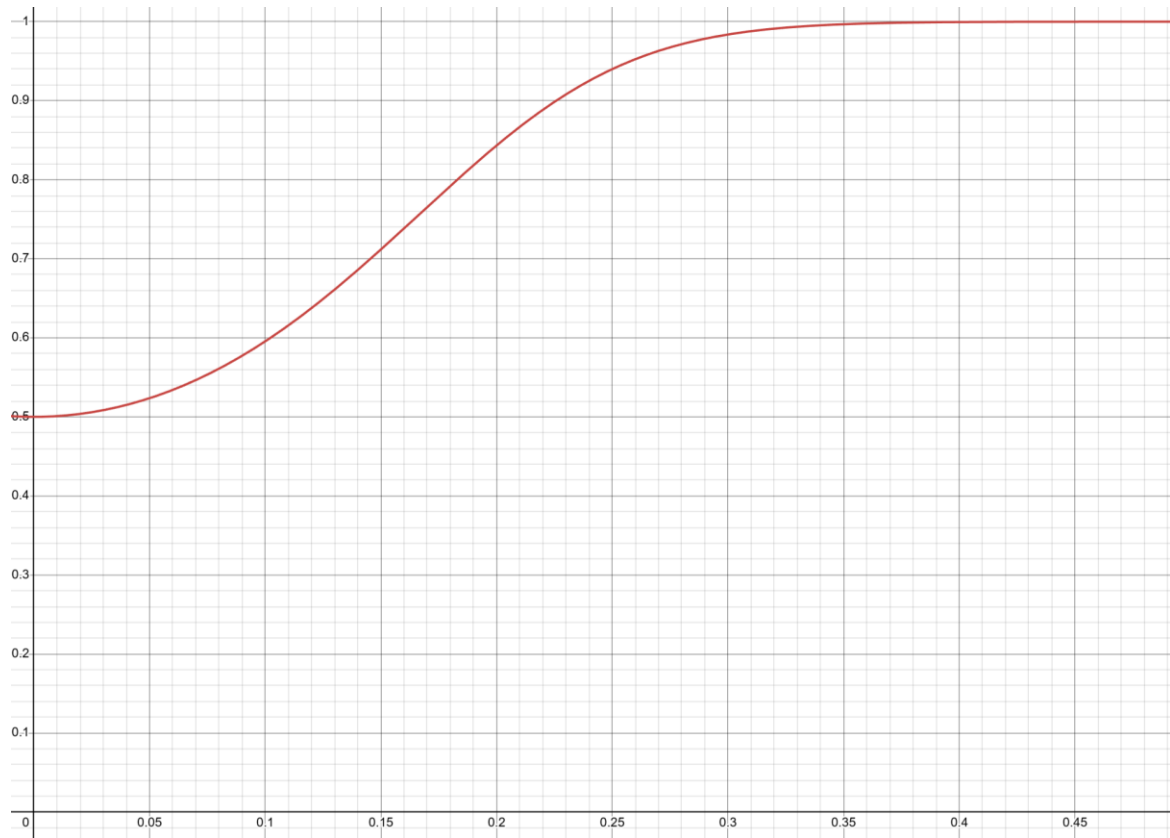


Figure 20: Illustration of the function used in the case of a competing logo. The y-axis represents the score between 0 and 1, and the x-axis represents the distance of the competing logo from the target logo.

4.2.1.2.4. Signal Integrity: Laplacian Variance

The final pillar of our quality index concerns visual signal integrity, evaluated by Laplacian Variance. This metric acts as a standardized estimator of blur detection. Technically, it consists of applying a Laplacian operator (spatial second derivative) to the logo image to highlight contours and rapid intensity transitions. The variance of this result quantifies the overall sharpness: a blurry image, poor in high frequencies, will produce a low variance, while a sharp image rich in detail will generate a high variance.

In a media valuation logic, the relationship between sharpness and value follows a threshold dynamic: below a certain clarity, the logo is unreadable and its marketing value is zero; conversely, once sufficient sharpness is reached, the increase in resolution no longer brings marginal cognitive gain to the spectator (saturation effect). To model this binary behavior (readable/unreadable), we reiterate the use of a sigmoid transfer function [formula]. The calibration of the parameters relies here on empirical image processing standards rather than on data distribution. We have fixed the central value at 100, a threshold generally accepted as the switch to acceptable sharpness, and defined a slope of $k=0.05$. This setting ensures a rapid transition to the maximum

score (1.0), reflecting the fact that once readability is guaranteed, the "surplus" of sharpness does not generate additional economic value

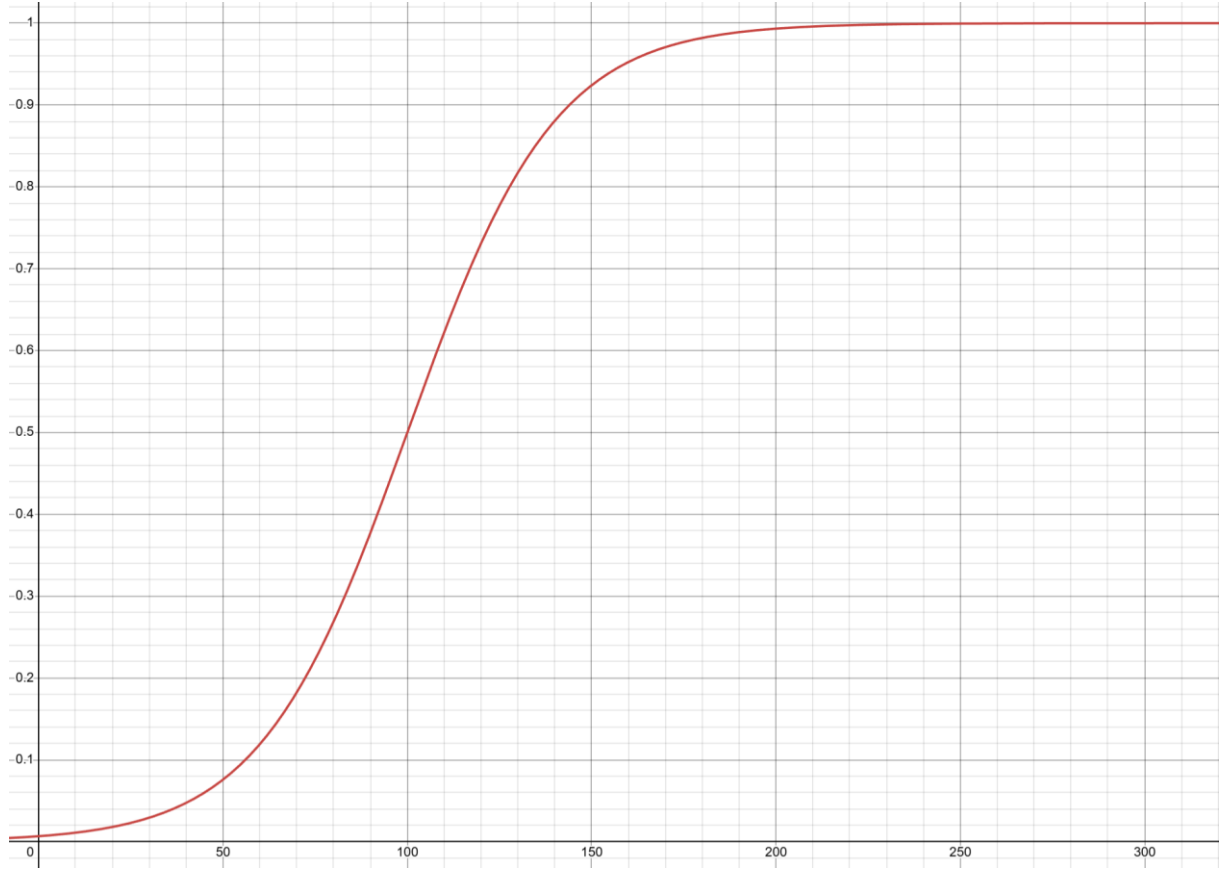


Figure 21: Illustration of the function used. The y-axis represents the score between 0 and 1, and the x-axis represents the Laplacian variance value.

4.2.1.3. Final Equation

The final valuation equation results from the multiplicative aggregation of all the quality factors previously defined, weighted by the audience volume and the exposure duration. For a logo i identified at time t , the instantaneous value is modeled by the product of the reference Cost per View, the display time, and the composite quality index (divided by 5 to obtain the value for one frame):

$$V_{i,t} = N_{vues} \cdot CPV_{sec} \cdot \frac{1}{5} \cdot QI_{i,t}$$

Where :

$$QI_{i,t} = \underbrace{\left[\frac{1}{1 + e^{-100(Area_{i,t} - 0.015)}} \right]}_{\text{Taille}} \cdot \underbrace{\left[e^{-\frac{d_{i,t}^2}{0.08}} \right]}_{\text{Distance}} \cdot \underbrace{\left[\frac{1}{1 + e^{-0.05(Lap_{i,t} - 100)}} \right]}_{\text{Netteté}} \cdot \underbrace{\sqrt{\frac{Area_{i,t}}{Area_{i,t} + \sum_{j \neq i} (Area_{j,t} \cdot e^{-\frac{d_{i,j,t}^2}{0.02}})}}}_{\text{Encombrement}}$$

By extension, the global valuation of a match is obtained by the temporal summation of these instantaneous values over all the logos detected during the video sequence, allowing for a cumulative financial value for the event:

$$Value_{Total} = \sum_t \sum_i V_{i,t}$$

4.2.2. Towards advanced modeling: critical discussion

We propose here an alternative theoretical modeling, conceptually more robust although technically more complex to implement. It is important to note that this formulation was not deployed within the framework of this prototype; the primary purpose of this section is to feed the critical discussion on the limits of the current model and to outline the prospects for evolution toward a finer measurement of media value.

4.2.2.1. Measuring engagement: a multimodal approach

The major limitation of the previous equation lies in its hypothesis of engagement linearity: it presupposes uniform spectator attention, independent of the dramatic intensity of the match or the sequence. However, this hypothesis constitutes a strong approximation. Marketing literature suggests that a spectator emotionally engaged in an action presents an increased permeability to peripheral advertising stimuli. It therefore appears relevant to integrate a dynamic multiplier coefficient reflecting this variable state of engagement. We propose two exploratory approaches to quantify this phenomenon. Although these metrics require subsequent empirical calibration on real data, they offer a solid conceptual framework for contextual valuation.

4.2.2.1.1. Social Amplification Factor: Sentiment Analysis

Our initial modeling relied exclusively on passive broadcast metrics (number of views), obscuring the richness of social metadata inherent to platforms like YouTube. However, Likes and comments constitute active engagement indicators that are far more relevant for measuring the community's real investment. The act of commenting, requiring greater cognitive and temporal effort than a simple click, signals a maximum level of interest. A high ratio between cumulative interactions and the number of views thus reflects a captive audience (engaged "Fan Base") rather than volatile consumption.

To refine this quantitative measurement, we propose integrating a qualitative layer via Natural Language Processing (NLP). Sentiment analysis applied to the corpus of comments allows for the evaluation of the emotional tone of the reception (enthusiasm, controversy, indifference). We therefore define a Social Amplification Factor whose formulation mechanically grants superior weight to comments compared to "Likes," in order to reward deep engagement. This metric allows for the modulation of the final value by taking into account the heterogeneity of interest shown in each match:

$$M_{engagement} = 1 + w_1 \cdot Sentiment_{avg} + w_2 \cdot \left(\frac{Like + w_3 \cdot Comment}{Vues} \right)$$

4.2.2.1.2. Contextual Intensity Detection: Audio Signal Analysis

The second methodological approach exploits the audio track, inseparable from the video stream, as a vector for detecting intra-match dramatic intensity. There is indeed a strong correlation between ambient sound volume (exclamations of commentators, crowd reaction) and the spectacular nature of a game action. These peaks in sound intensity signal moments that maximize the spectator's visual attention and increase the probability of rewinding to watch the action again.

To objectify this phenomenon, we propose a formula based on audio signal analysis. The algorithm compares the instantaneous sound amplitude to the smoothed average intensity over the duration of the video. When a significant deviation from the average is detected (decibel peak), it triggers a value multiplier for the logos exposed during this critical interval. This dynamic approach corrects the linearity flaw of the previous model by recognizing that attentional engagement is not constant, but fluctuates with the rhythm of the match:

$$M_{audio}(t) = 1 + w_4 \cdot \max \left(0, \frac{dB(t) - (1 + w_5) \cdot \overline{dB}}{\overline{dB}} \right)$$

4.2.2.2. Habituation Modeling

The major structural bias of our current model lies in the linearity of temporal valuation. By attributing a constant value to each additional second of exposure, we ignore a fundamental principle of cognitive psychology: habituation. Human attention is not an inexhaustible resource; it saturates. Once a visual stimulus (the logo) has been detected and processed by the brain, its prolonged maintenance on the screen does

not generate a proportional gain in memorization. On the contrary, the spectator eventually treats this stimulus as background information, reducing its marketing impact.

To correct this bias, it would be imperative to apply a function of diminishing marginal returns to the duration of exposure. Specifically, the first few seconds of appearance, those of "discovery," should be financially overweighted, while the following seconds would see their unit value gradually decrease until reaching a plateau, reflecting the saturation of attention. Of course, the global quality score should be taken into account in all of this and not just the logo's exposure time; if the logo is small and far from the action during the first 10 seconds, it would be unfair to consider that the spectator has been fully exposed to it for 10 seconds. We therefore propose the following formula:

$$F_{duration}(t) = e^{-w_6 \cdot \sum_{k=0}^{t-1} (QI_{i,k} \cdot \frac{1}{5})}$$

4.2.2.3. Spatial Centrality: Adaptive Gaussian Decay

Finally, our current model assumes a rigidity of attention (constant sigma) that does not withstand the analysis of game dynamics. The width of the attentional focus varies considerably depending on the phase of play. During an offensive buildup sequence (set play, multiple screens), the spectator's attention is diffuse and panoramic, requiring a high sigma to value peripheral logos. Conversely, during a long-distance shot, the gaze locks onto the ballistic trajectory of the ball (tunnel effect), justifying a very low sigma and an increased penalty for off-axis elements.

Furthermore, this adaptability should extend to the very nature of the brand. Cognitive familiarity plays a key role in peripheral vision. An iconic brand (Nike, for example) benefits from immediate recognition even at low resolution or in lateral vision, thanks to pre-established recognition heuristics. In contrast, an emerging brand or one with complex graphics requires direct foveal fixation to be decoded. Treating these two cases with an identical decay curve constitutes an economic inefficiency.

We therefore suggest the introduction of adaptive multiplier coefficients for the sigma parameter, modulating spatial tolerance based on the type of action (context) and the brand awareness (stimulus). Although complex to generalize on a large scale, this tailor-made approach would be technically possible for a personalized analysis.

$$f(x) = e^{-\frac{x^2}{2(\sigma_{base} \cdot w_7 \cdot w_8)^2}}$$

4.2.2.4. Improved Final Equation

The synthesis of these conceptual improvements results in a dynamic valuation equation, significantly more granular than the initial linear model. For a logo i at time t , the instantaneous value $v_{i,t}$ no longer depends solely on its physical visibility, but on its contextual and cognitive resonance. It is defined as the product of the reference cost, adaptive visual quality metrics (including a variable sigma), engagement multipliers (audio and social), and the decay factor linked to habituation:

$$V_{i,t} = N_{vues} \cdot CPV_{sec} \cdot \frac{1}{5} \cdot QI_{i,t} \cdot \underbrace{M_{audio}(t)}_{\text{Intensité}} \cdot \underbrace{F_{duration}(t)}_{\text{Saturation}}$$

Where :

$$QI_{i,t} = \underbrace{\left[\frac{1}{1 + e^{-100(Area_{i,t} - 0.015)}} \right]}_{\text{Taille}} \cdot \underbrace{\left[e^{-\frac{d_{i,t}^2}{2(\sigma_{base}^{w_7 w_8})^2}} \right]}_{\text{Distance Adaptative}} \cdot \underbrace{\left[\frac{1}{1 + e^{-0.05(Lap_{i,t} - 100)}} \right]}_{\text{Netteté}} \cdot \underbrace{\sqrt{\frac{Area_{i,t}}{Area_{i,t} + \sum_{j \neq i} (Area_{j,t} \cdot e^{-\frac{d_{i,j,t}^2}{0.02}})}}}_{\text{Encombrement}}$$

Consequently, the total valuation generated by a match is obtained by the temporal integration of these instantaneous values for all detected logos:

$$Value_{Totale} = M_{engagement} \times \sum_t \sum_i V_{i,t}$$

With:

- w_1 : The impact of sentiment (whether comments are positive or negative).
- w_2 : The global importance of interactions (likes and comments) relative to views.
- w_3 : The specific multiplier for comments (as a comment is worth more than a like).
- w_4 : The strength of the bonus granted during a sound peak (the intensity of the premium).
- w_5 : The trigger threshold (the percentage of volume above the average necessary to activate the bonus).
- w_6 : The saturation speed (how quickly the value decreases when the logo stays on screen too long).
- w_7 : The adjustment according to the game phase (wide focus during buildup, narrow focus during a shot).
- w_8 : The advantage linked to brand awareness (ease of recognition in peripheral vision)

5. Quantification and Prediction of Media Value

The formalization of the econometric model established in the previous section now allows us to convert raw exposure metrics into a tangible media value, weighted by the actual audience. With this methodological foundation in place, we can move to the operational phase of data exploitation to address the two strategic objectives of this project. The first part consists of extracting analytical insights from the historical corpus to understand the current valuation drivers. The second part aims to go beyond retrospective analysis to exploit this data history in order to build a predictive model capable of anticipating the economic potential of a future matchup.

5.1. Descriptive Analysis

The descriptive analysis constitutes the first fundamental step of this quantitative exploitation. The objective is to extract the rich information from the dataset to identify levers of value creation: which teams dominate the media space, which zones capture the most attention, and how these performances compare to traditional sporting hierarchies. Although the granularity of our data theoretically allows us to drill down to action-by-action or player-by-player analysis, we will focus here on a macroscopic view by team and by match.

Over the entire observed period, which covers 522 matches across 76 days, the total valuation of exposure reached \$14,607,335 USD. This represents an average value creation of approximately \$200,000 USD per day and, more specifically, nearly \$28,000 USD per match. However, these global figures mask deep disparities; the distribution of value is heavily concentrated among a handful of franchises capable of generating massive virality.

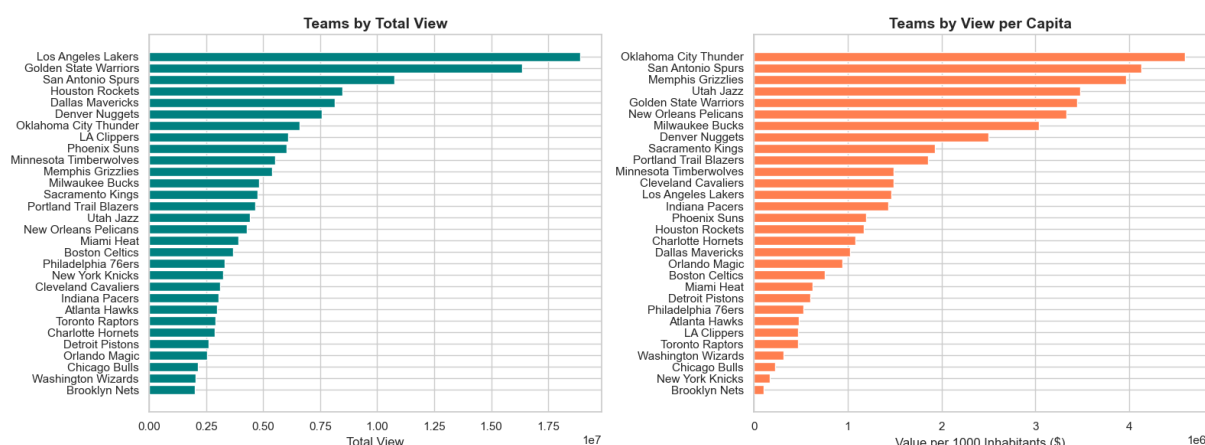
The analysis logically begins with raw audience volume, which forms the bedrock of value creation. The ranking by number of views (see "Teams by Total View" chart) highlights an unchallenged dominance by the Western Conference, which monopolizes the top ten spots. The Los Angeles Lakers and Golden State Warriors sit at the summit, capturing the bulk of global attention thanks to their respective superstars: LeBron James, Stephen Curry, and Victor Wembanyama. Conversely, the Eastern Conference suffers from a notable exposure deficit, with its top representative, the Milwaukee Bucks, appearing only in eleventh position.

The conversion of this audience into monetary value generally maintains this hierarchy, confirming that view volume remains the primary driver of the "Shadow Media" economy. The Lakers and Warriors widen the gap, generating over \$3.5 million and \$3 million respectively, far ahead of the rest of the league. However, when this performance is related to market size (see "Value per Capita" chart), the perspective

changes radically. Certain "small markets" like Memphis or Oklahoma City dominate this efficiency ranking. While this performance is understandable for OKC— unquestionably the best team of the current season—it is more surprising for the Grizzlies, who are at the bottom of the standings. This demonstrates that raw sporting performance is not the only lever: the presence of spectacular players generating viral highlights is just as crucial for capturing the interest of an audience that sometimes follows individuals more than franchises.

The scatterplot confirms this disconnection between local demographics and digital media power. While the Lakers manage to align their market size with exceptional value, franchises located in massive metropolises, such as the New York Knicks or Brooklyn Nets, massively underperform relative to a population base of nearly 20 million inhabitants. This finding validates the hypothesis that YouTube consumption follows a logic of "Star Power" and globalized storytelling, breaking free from the geographical boundaries that traditionally govern ticketing revenue.

It is, however, crucial to emphasize that the relationship between the number of views and generated value is not strictly linear, as subtle variations appear depending on the team. Indeed, each franchise plays half of its matches at home, in an arena where it controls the commercial configuration. The final value therefore intrinsically depends on the quality of the advertising inventory: sponsor density and logo size. This structural nuance is visible when comparing the Phoenix Suns and the Oklahoma City Thunder: although the Thunder accumulate a higher volume of views, the Suns manage to generate a higher total value. This suggests that the Phoenix arena offers better visibility or a more lucrative layout for its partners, thus maximizing the return on every view generated



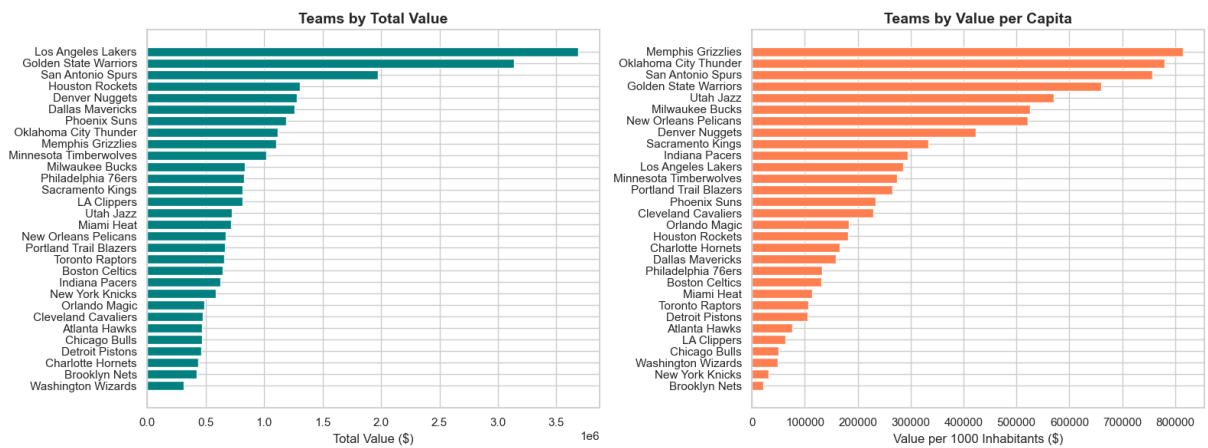


Figure 22: Ranking of the teams generating the most views and values, put into perspective with their market size.

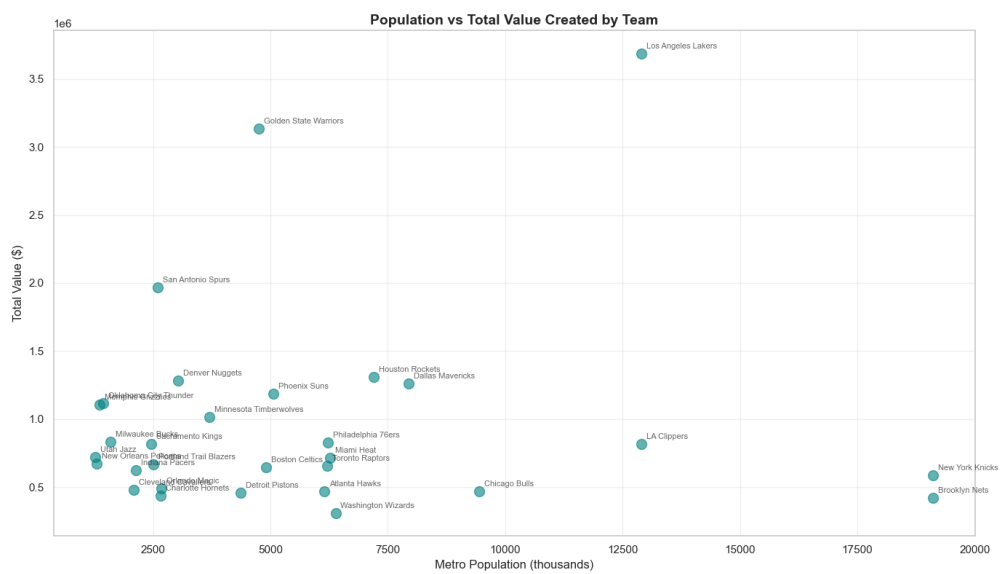


Figure 23: Scatterplot visualizing the relationship between market size and the value generated.

It is now relevant to refine our analysis by descending to the scale of exposure zones. Indeed, as explained, the media value generated does not depend solely on the team's popularity, but also on the visual configuration of the court: the number of sponsors, their physical size, and above all, their placement relative to the camera angles preferred by broadcasters play a crucial role.

The examination of cumulative totals reveals a clear hierarchy of advertising spaces: the "Back-Court Logo" zone (located near the baseline) establishes itself as the most lucrative location, with a total valuation exceeding \$4.3 million over the period. It is followed by the "Mid-Court Logo" (\$3.3 million), while lateral zones such as the "Side-Court LED" and the "Basket Logo" generate more modest volumes, oscillating between \$2.2 and \$2.7 million.



Figure 24: Breakdown of the total and average value generated during the current season for each team.

To understand these fluctuations, it is necessary to decompose the Quality Index. The analysis of the components reveals that the dominance of the "Back-Court Logo" is explained by an optimal combination: this zone benefits from a significant physical size (Size Score often above 0.50, peaking at 0.68 for the Boston Celtics) and an exceptional "Share of Voice" (often above 0.90), indicating that the logo is generally isolated and faces little visual competition.

In contrast, the "Basket Logo" presents an interesting paradox: although it achieves the best readability scores due to its sharpness (close to 1.00 for almost all teams), its small size (scores around 0.15) and the visual clutter of the basket structure (low Share of Voice around 0.40) drag down its overall quality score, which stagnates at a very low level (approximately 0.02-0.03).

Finally, the analysis of exposure time highlights major structural disparities that penalize certain franchises. While the majority of teams show an exposure rate for the "Back-Court Logo" exceeding 75%, some exceptions are striking: the Houston Rockets (10.7%), the New Orleans Pelicans (2.2%), and the Sacramento Kings (7.3%) show nearly zero exposure rates in this strategic zone. This is, in reality, quite simply due to the fact that these teams do not possess a back-court logo; the same could be said for the side-court logo, which is not present on all courts, or the mid-court logo, which is not the same size on all courts, explaining the difference in exposure percentage.

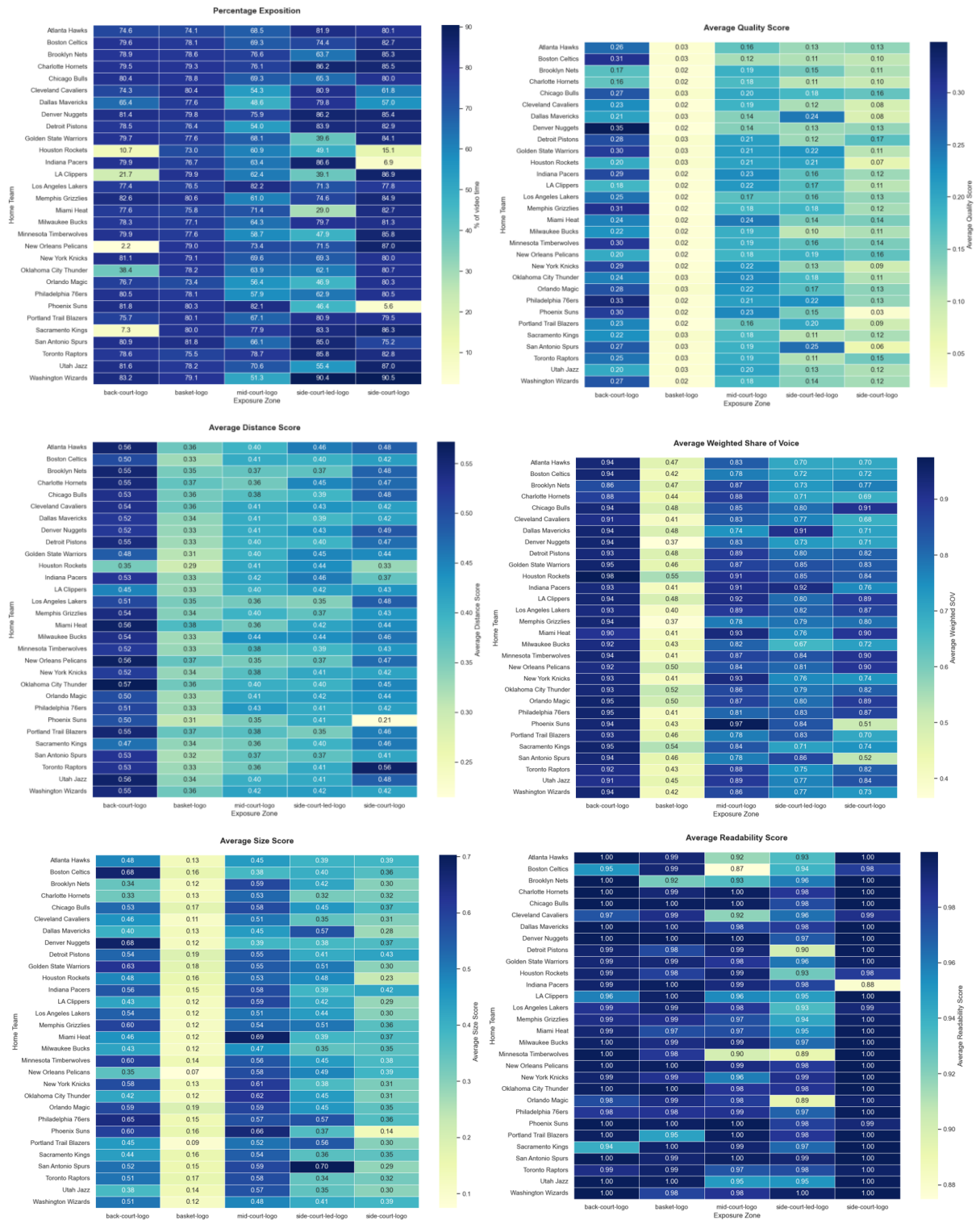
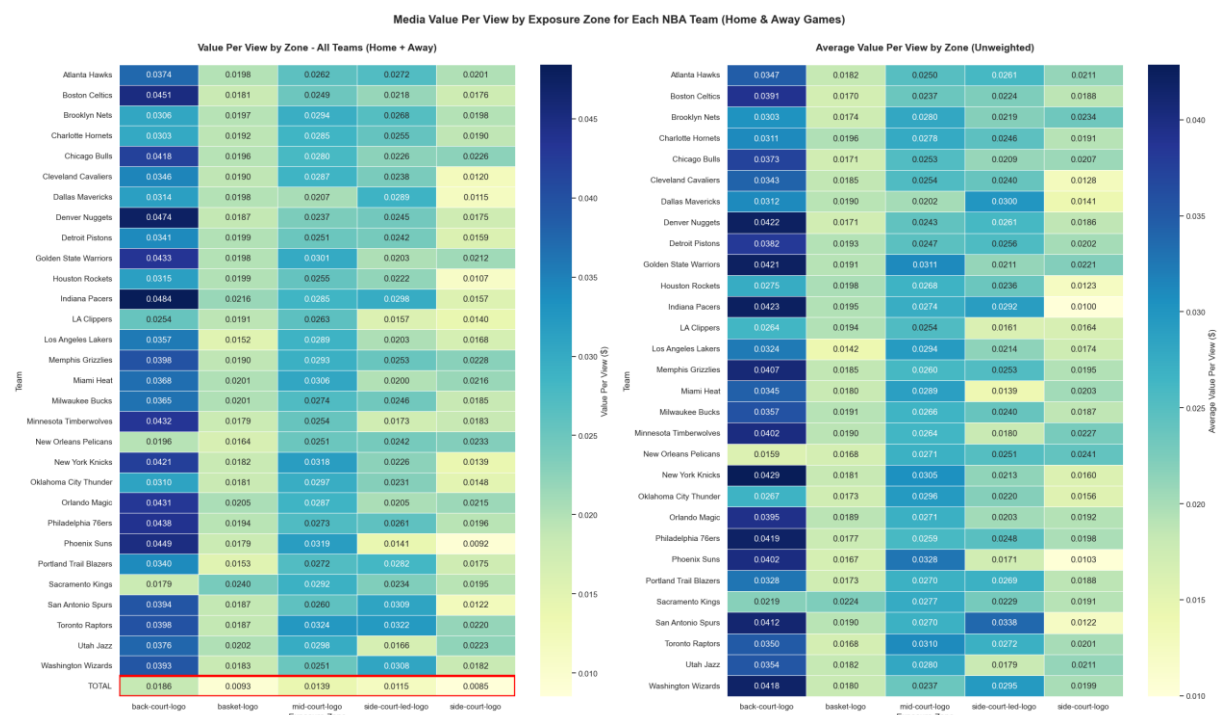


Figure 25: Breakdown of the different scores composing the quality index for each court to understand the disparities in value generated.

All this explains the significant variations in yield per view (Value Per View) for each team and each zone. By neutralizing audience volume to focus solely on the intrinsic

efficiency of the advertising medium, it is clear that the "Back-Court" zone remains the queen of profitability with an average of \$0.0186 generated per view—more than double that of the "Basket Logo" (\$0.0093) or the "Side-Court Logo" (\$0.0085). This graph allows for the identification of structural "top performers": the Indiana Pacers and the Denver Nuggets, for example, show exceptional yields on their baseline (respectively \$0.0484 and \$0.0474 per view), maximizing every gaze directed at their games.



In synthesis, this analysis highlights the significant economic scale of unofficial highlights in the NBA. The major takeaway lies in the disconnection between market size (city) and the digital value generated: on YouTube, it is not local demographics that prevail, but "Star Power" and the spectacular nature of the game. Small-market franchises like Memphis or San Antonio thus manage to rival historical giants thanks to emblematic figures like Ja Morant or Victor Wembanyama, who capture a globalized audience.

For stakeholders in the sports ecosystem, these results offer an interesting strategic decision-making tool. For sponsors, this approach allows for the identification of undervalued assets by targeting not necessarily the most-viewed teams, but those offering the best "Value Per View" and the best structural visibility (quality of exposure). For franchises, the zone-by-zone analysis reveals concrete optimization levers: resizing a logo or slightly adjusting camera angles can mechanically increase the valuation of their advertising inventory.

5.2. Predictive Modeling

Initially, the objective was not to predict the value generated by a match, but rather its total number of views, as this variable depends on fewer contextual factors. However, the analysis shows that these two metrics are strongly and linearly correlated. This correlation is logical since the media value created is, ultimately, a multiplier of the audience volume. Consequently, we chose the value generated per match as the target variable for our modeling. The calculation process is carried out in two steps: we first sum the value of each detected zone to obtain the total value of a video. Then, we add up the values of all videos associated with the same matchup to determine the total value generated by the match, thus constituting our Y variable.

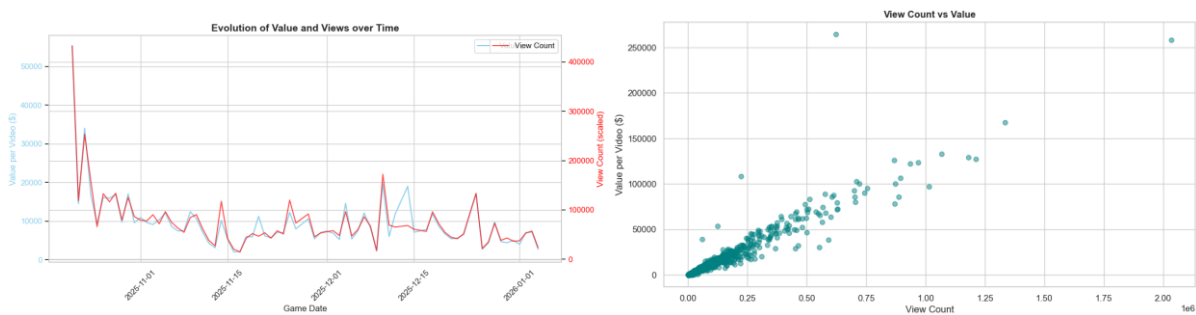


Figure 26: Relationship between the value generated and the number of views.

The construction of the feature space, or Feature Engineering, relies on the exhaustive history of previous seasons. Although we have granularity down to the individual player level, we chose to focus on team metrics to ensure the statistical robustness of the model. This work allowed for the generation of relevant synthetic variables, such as moving averages, streak dynamics (wins or losses), and the pre-match win percentage as an indicator of the balance of power. We also integrated the density of the daily schedule to test the hypothesis of attention dilution: if many games take place simultaneously, the audience disperses, while a reduced schedule encourages a concentration of spectators on the few available games.

To refine these predictions, we created moving average variables for both the home and away teams, including audience history over the last five games as well as the teams' global statistics over that same period, such as points scored or the point spread with the opponent. Weekend temporality and sports performance indicators, such as win percentage and result streaks, were also added. After a data normalization phase, three architectures were put into competition: a Linear Regression, a Random Forest, and an XGBoost. This methodological choice allows for exploring the trade-off between simplicity of interpretation and computational power.

The objective of this competition is to propose two complementary solutions. The first relies on Linear Regression for its mathematical transparency, while the second

exploits more complex models, such as Random Forest and XGBoost, to maximize predictive performance. This dual approach allows us to discuss the final choice of the model based on business needs, weighing raw precision against the ability to clearly explain the drivers of media value.

5.2.1. Linear Regression

For the development of this first model, several subsets of variables were tested to identify their predictive potential. The final variables selected include the average viewership of the home and away teams over their last five games, as well as the timing of the match to distinguish games taking place during the weekend. The model also integrates the total number of games scheduled on the same day to measure the impact of media competition, as well as the respective win percentages of both teams before tip-off. Finally, indicators of recent sports performance complete this dataset, notably the average point differential (plus/minus) and the average points scored by each team during their last five game.

Results for: Linear Regression	
Metric	Value
MAE	: \$11,034.47
RMSE	: \$16,069.95
R ²	: 0.5833

Figure 27: Linear regression results.

The results of the linear regression show moderate performance in predicting the media value per match. The coefficient of determination of 0.5833 indicates that the model manages to explain approximately 58% of the data variance, which shows a capacity to capture global trends but leaves a significant amount of uncertainty. This imprecision is reflected in the Mean Absolute Error (MAE), which amounts to \$11,034.

Putting these figures into perspective with the average generated value of \$23,000 per match, it can be observed that the average error represents nearly 48% of the typical value of a game. Such a discrepancy suggests that the model struggles to model the complexity of factors influencing valuation, particularly for matches whose audiences deviate from the norm. The significant gap between the MAE and the RMSE also indicates the presence of outliers or major prediction errors on certain specific events, which the rigid structure of linear regression fails to absorb.

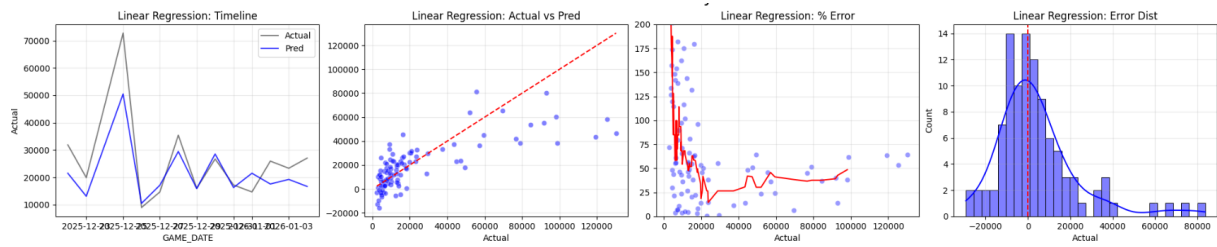


Figure 28: Error analysis of the linear regression.

Detailed analysis of the error analysis graphs allows for a nuanced view of the initial global impressions. Although the first graph (Timeline) seems to show a model capable of following general trends, the second graph (Actual vs Pred) reveals that the model struggles to capture the full amplitude of high values, with many points deviating significantly from the ideal line due to a systematic underestimation of high-value matches.

The study of the relative error distribution (Graph 3) is even more critical: it demonstrates major instability for low viewership, where a majority of predictions deviate by more than 50% from reality, with error peaks exceeding 150%. This high volatility is explained by the linear model's difficulty in managing the statistical "noise" inherent in small volumes.

The model's reliability only becomes somewhat more acceptable for the high-exposure match segment, where the red curve of the average error stabilizes, even if the results remain generally imprecise. Finally, the fourth graph (Error Dist) confirms this trend with a distribution of residuals that, although centered on zero, presents a significant "tail" to the right, illustrating the persistence of massive prediction errors on part of the dataset.

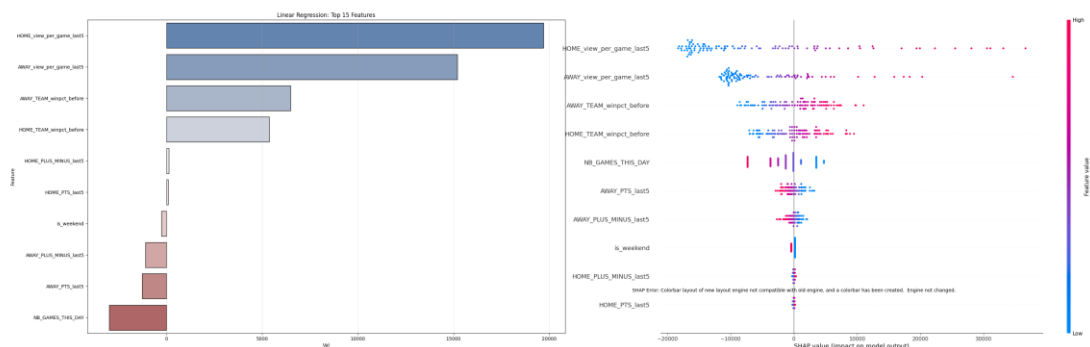


Figure 29: Interpretability of the linear regression.

The interpretability analysis of the linear regression highlights the dominant drivers that dictate the media value of a matchup. Unsurprisingly, the recent viewership history of

both teams constitutes the most powerful predictive factor. Variables related to the average number of views per match over the last five outings (for both the home and away teams) show the highest coefficients, confirming that immediate popularity and the "hype" surrounding franchises are the primary drivers of the value generated on YouTube.

The model also emphasizes the importance of sports competitiveness through pre-match win percentages. A high-performing team mechanically attracts more attention, which translates into a direct positive impact on media valuation. Conversely, the factor related to the number of games played on the same day (multiplex) acts as a drag: the busier the schedule, the more the spectator's attention is diluted, which tends to reduce the unit value of each individual game.

Finally, the SHAP graph allows for the observation of these impacts in a more granular way. It clearly shows that high values for past audiences (in red) systematically pull the prediction upward with a very marked amplitude. Although other factors such as point differential or weekend timing are integrated, their influence remains marginal compared to the overwhelming weight of the teams' notoriety, thus validating a market dynamic where the prestige of franchises prevails over the logistical conditions of the match.

5.2.2. Random Forest and XGBoost

To overcome the limitations of the linear approach, we implemented more complex architectures such as Random Forest and XGBoost. Unlike the previous model, these algorithms integrate additional variables that significantly improve overall predictive performance without benefiting the linear regression.

The feature space of these models is thus enriched with more granular indicators, such as the total number of losses accumulated by both teams before the matchup and their current result streaks (wins or losses). Finally, the home team ID was added using One-Hot encoding. This variable is of interest, and we only take the home team because the court on which the match is played is, as seen previously, important and affects the quantity and quality of exposure and thus the final value.

Results for: XGBoost	
Metric	Value
MAE	: \$8,892.91
RMSE	: \$14,467.93
R ²	: 0.6622

Results for: Random Forest	
Metric	Value
MAE	: \$9,461.43
RMSE	: \$14,838.65
R ²	: 0.6447

Figure 30: Random Forest and XGBoost results.

Ensemble models, represented here by Random Forest and XGBoost, mark a clear progression in the precision of media value predictions. XGBoost emerges as the most performing architecture with a coefficient of determination of 0.6622, followed closely by Random Forest at 0.6447. This improvement mechanically translates into a reduction in average errors: XGBoost brings the Mean Absolute Error (MAE) down to \$8,892.91, while Random Forest shows \$9,461.43.

Despite this increase in power, these models retain some of the instability observed previously. The Root Mean Square Error (RMSE), which stands at \$14,467.93 for XGBoost, remains high compared to the MAE. This confirms that even the most advanced non-linear algorithms still struggle to predict with total precision the "peaks" in value generated by exceptionally viral matchups, although they absorb the general complexity of the data better than the linear model.

Comparing these results with the initial linear regression (R^2 of 0.5833), the performance gain is undeniable. Moving to an ensemble model allows for a reduction of the MAE by more than \$2,000, representing an improvement of nearly 20% in average precision.

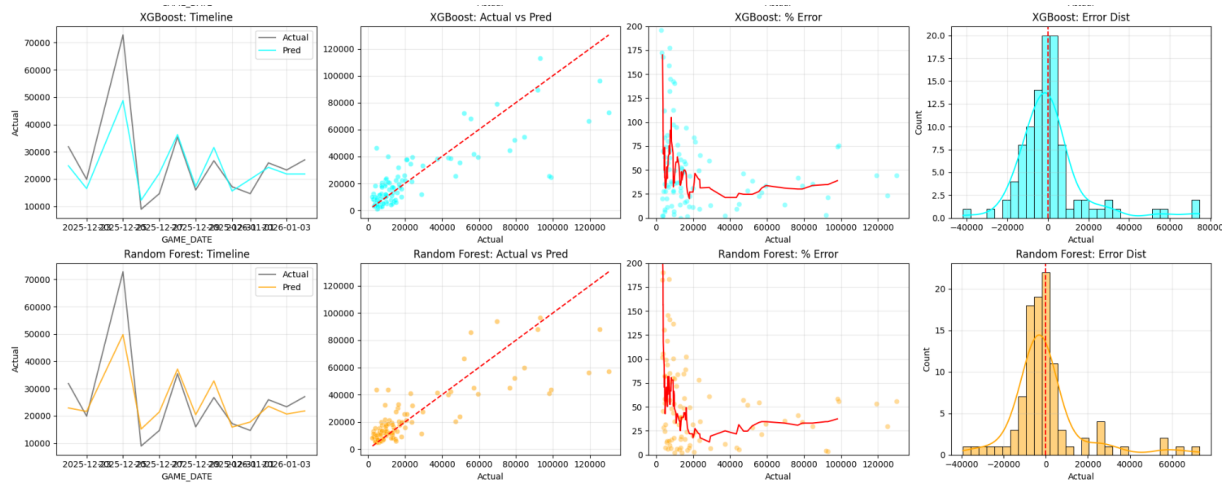


Figure 31: Error analysis of the Random Forest and XGBoost models.

The error analysis for these ensemble models presents trends generally similar to those observed for the linear regression, confirming the persistence of certain structural challenges in the data.

However, a slight improvement in stability is noted, illustrated by an error distribution (Error Dist) that is a bit more tightly centered around zero and a relative error curve (% Error) that drops more quickly for matches with moderate viewership. Nevertheless, the "Actual vs Pred" graph shows that even with XGBoost or Random Forest, the difficulty in accurately predicting extreme values persists, as the models continue to underestimate the matchups with the strongest viral impact.

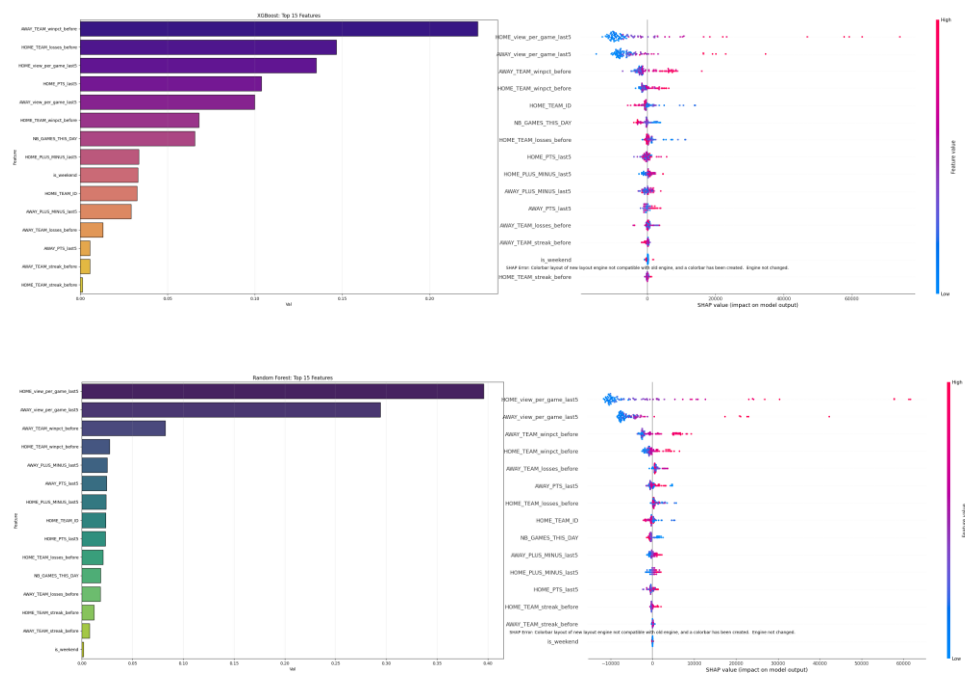


Figure 32: Interpretability of the Random Forest and XGBoost models.

The interpretability analysis of the Random Forest confirms the hierarchy of factors identified previously, while refining the understanding of non-linear interactions. As with the other models, the historical notoriety of the teams remains the main driver of media value, with past home and away viewership largely dominating the ranking of the most important variables.

However, the Random Forest brings an interesting nuance by granting more significant weight to immediate performance indicators, such as the pre-match win percentage and recent point differential. The SHAP graph reveals that while very high viewership (in red) guarantees high valuation, the model is capable of more finely modulating its predictions for matches with high sporting stakes, thus capturing value gains that the linear regression could not perceive.

5.2.3. Conclusion and Model Choice

The final choice illustrates the classic dilemma between the transparency of Linear Regression and the predictive power of XGBoost. While the latter is technically selected for its ability to reduce average error and better capture the non-linearities of virality, this superiority remains relative. In reality, the error analysis indicates that neither of the two solutions is currently robust enough for industrial production or to provide truly actionable insights. The instability of the predictions necessitates a substantial enrichment of the data before being able to guarantee the financial reliability required.

6. Conclusion

6.1. Synthesis

This research project has focused on resolving a central issue in the modern sports economy: the objectification and financial valuation of "Shadow Media." We have demonstrated that unofficial NBA highlights on YouTube, far from being a total loss for the league, constitute a considerable source of exposure value for sponsors.

To capture this invisible value, we deployed a hybrid methodology at the intersection of Computer Vision and marketing. On the technical side, the implementation of a YOLOv11-OB (Oriented Bounding Box) architecture proved decisive. This choice allowed for robust handling of the complex geometric constraints of NBA courts (perspective, rotations). The rigorous creation of a 2,000-image dataset, refined by a Human-in-the-Loop strategy and iterative enrichment, achieved a detection accuracy (mAP50) of 0.95, validating the reliability of the extraction pipeline.

Beyond simple detection, the strength of this project lies in its economic modeling proposal. By moving beyond the raw metric of screen time to integrate qualitative criteria (saliency, centrality, sharpness, share of voice), we built an Exposure Quality indicator that is as faithful as possible to the reality of cognitive impact. Coupled with predictive audience analysis, this approach offers decision-makers (franchises and advertisers) a concrete Business Intelligence tool to audit the actual performance of their marketing assets in the viral digital ecosystem.

6.2. Biases and Limitations of the Current Mode

Although the realization of this project has been a major learning and research experience, it presents intrinsic imperfections.

6.2.1. The model

The greatest progress to be made lies in the model architecture and the way the problem is approached. In this project, we treated the subject as an object detection problem. This is sufficient to a certain extent, but it remains a naive approach that leads to a loss of information. Typically, the model processes images sequentially but independently: it does not make the link between frame A and frame B. Therefore, we lose continuity information: we know there is a logo, but the model cannot tell if it is the same one as before, and the same applies to the ball.

This limitation leads us to count a logo in the formula even if it appears on screen for only a single frame or a dozen consecutive frames, whereas for the human eye, it is almost imperceptible. To solve this problem, we should not settle for just detecting the logo, but track it using object tracking models. Obviously, this would require revising the database, but it would allow for a much fairer evaluation and the retrieval of information on successive appearance time. For equivalent exposure time, a logo that appears in continuous 3-second blocks has a much higher chance of being seen and remembered than a logo that appears in fragments of less than a second.

6.2.2. The data

As it stands, the database is correct in terms of volume for this specific problem, but the major criticism for a more advanced project concerns data granularity. Currently, we settle for detecting fairly wide "zones" rather than precise brands. Typically, we detect the global LED panel and not the presence of a specific logo on that panel. This is a difficult task because it is not static: there are animations and sometimes several brands on the same panel.

Similarly, the `basket_logo` class is too generic: a basket often features approximately 5 different logos and sometimes even an LED panel installed on the structure, which poses the same distinction problems. Finally, we only distinguished 5 zones, totally omitting jersey patches and other LED panels in the arena. For our specific case focusing on highlights, this was not critical as these elements are rarely visible, but for an extension of the project, this is a point that must absolutely be taken into account.

It should also be noted that the data does not take into account the obstruction of logos by players, which happens often. Although they often remain visible and understandable with one or two players in front, they are not as clear as a logo with no one in front or on top of it, and therefore the exposure quality is not the same.

Furthermore, the single-step data collection strategy poses a sustainability issue given the volatility of the YouTube platform. Over time, certain videos or channels may be deleted for various reasons, which already seems to be occurring in our observations. This fragility of the digital corpus can limit the exhaustiveness of the analyses and highlights the need to implement more regular collection and storage pipelines to secure the data history.

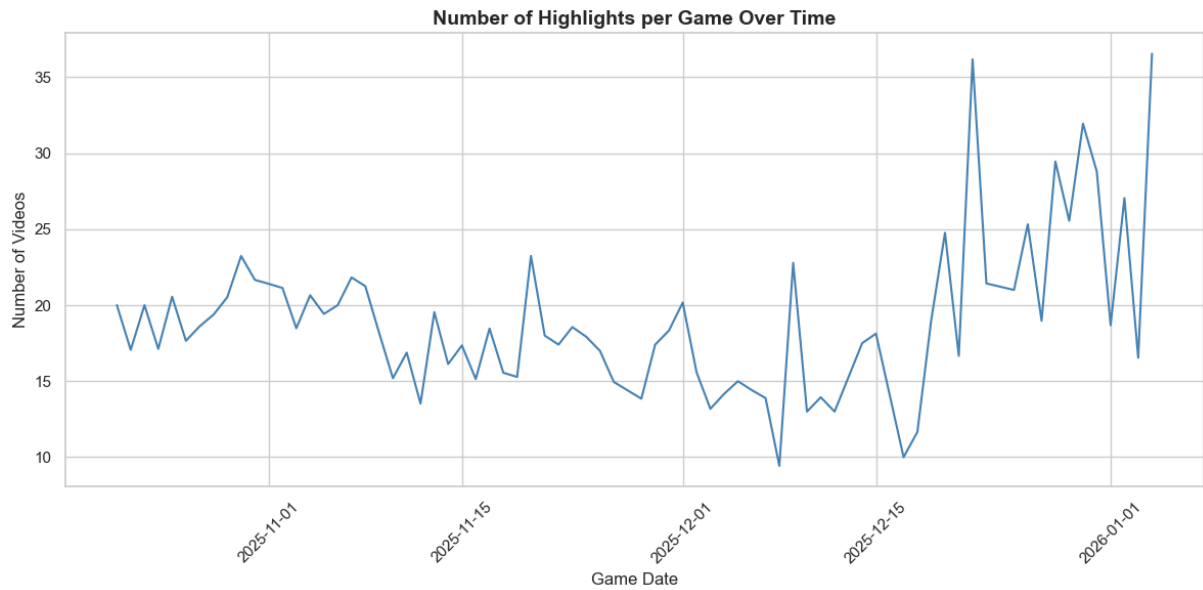


Figure 33 : Time evolution of the number of videos found on YouTube.

6.2.3. The Valuation Formula

This point was already mentioned during the presentation of the formulas, but it is important to reiterate: having a fairer formula, which takes into account the context and actual attention, is essential to have full confidence in the displayed figures.

6.3. Model Generalization

This final part opens the discussion on the scalability and the future of the project. We analyze here the opportunities for extending the model along two distinct axes: the diversification of sources within the existing NBA ecosystem ("In League") and the transposability of the architecture to other championships or sports disciplines ("Out League").

6.3.1. In-League Extension

Currently, the model is hyper-specialized in standard YouTube highlights, characterized by a massive predominance (99%) of wide lateral shots. If we wish to generalize the tool to all NBA content, it is imperative to enrich the training dataset with an even greater diversity of camera angles. Indeed, the current model shows its limits on tight zooms or ground-level shots that induce strong perspective distortions of the

logos. This is a minor technical obstacle that can be overcome by annotating more targeted data.

At the same time, this extension toward new content formats requires a revision of the valuation formula. On one hand, the Cost per View (CPV) must be dynamically recalibrated to reflect the economic reality of each broadcasting platform. On the other hand, the geometric parameters of the quality function must be adapted: for example, the analysis of vertical format videos (such as TikTok or Reels) would require modifying the sigma value in our attention modeling, as the gaze dynamics and image composition are radically different from the classic landscape format.

6.3.2. Out-League Extension

As it stands, the model cannot generalize to other leagues without retraining. Tests conducted on the G-League revealed significant failures: the model systematically confuses the institutional logos of the teams (at center court and in the backcourt) with commercial sponsors. This is a logical behavior: it has learned to ignore NBA team logos but does not possess this knowledge for other franchises. Furthermore, the zone classification becomes obsolete as soon as the court standard changes. Applying this model to the Betclic Élite or another sport would make no sense because the topology of advertising zones differs entirely.

However, while the model is not transferable as is, the "core" of the project—namely the detection architecture and the valuation pipeline is perfectly agnostic. The blockage is not structural; it is purely a data problem. To adapt the solution to a specific new league, it would suffice to build a new set of annotated data specific to that championship. This can take more or less time depending on the complexity of the task.

7. References

7.1. Market Analysis and Specialized Press

- **Zeng, A. (2025).** *Thesis on NBA media dynamics*. MIT Sloan.
<https://dspace.mit.edu/bitstream/handle/1721.1/163331/zeng-arzeng-msms-sloan-2025-thesis-not-signed.pdf>
- **SponsorUnited (2023).** *NBA Sponsorship Revenue & Spend Study 2022-23*.
<https://www.sportspro.com/news/sponsorship-marketing/nba-sponsorship-revenue-spend-2022-23-sponsorunited-study/>
- **Sportico (2025).** *How NBA teams and owners make money*.
<https://www.sportico.com/leagues/basketball/2025/how-nba-teams-owners-make-money-1234874170/>
- **Awful Announcing.** *FreeDawkins YouTube channel and NBA videos disappearance*.
<https://awfulannouncing.com/nba/freedawkins-youtube-channel-nba-videos-vanish.html>
- **Investment Markets.** *Sports Investing: A Rising Global Opportunity*.
<https://www.investmentmarkets.com.au/articles/entertainment-sports-gaming/sports-investing-a-rising-global-opportunity-296>

7.2. Advertising Costs (CPV) and Sponsorship Data

- **Simulmedia.** *How much do TV ads cost?*
<https://www.simulmedia.com/blog/how-much-do-tv-ads-cost>
- **MediaRadar.** *NBA Playoff Ad Spend Analysis*.
<https://www.mediaradar.com/blog/nba-playoff-ad-spend-surges-mediadar-unpacks-the-big-wins-for-abc-tnt-and-espn>
- **Adopter Media.** *YouTube Sponsorship Guide*.
<https://adopter.media/youtube-sponsorship-guide/>
- **Analyst Economy.** *Getting more out of media values*.
<https://analysteconomy.com/newsletter/f/%F0%9F%93%8A-getting-more-out-of-media-values>

7.3. Recommendations

- Clear explanation of the global NBA economy.
<https://www.youtube.com/watch?v=-Mj-J77C4Z0>

It should be noted that a multitude of other sources were consulted throughout the project to fully grasp the subject and verify certain figures. However, the references listed above constitute the primary sources used directly for the writing of this report and the construction of the valuation methodology.

8. Appendices

8.1. Methodology for the collection of URLs of interest

The methodology adopted to collect YouTube URLs of interest, which will then serve to quantify exposure, is as follows.

First, we rely on game data previously collected via the NBA API. Once this information is consolidated (home team, away team, and game date), we construct a search query modeled after the titling standards observed on YouTube according to the following schema:

```
Search_query = df['HOME_TEAM_NAME'] + ' vs ' + df['AWAY_TEAM_NAME'] + ' Full  
game highlights | ' + df['game_date_str'] + ' NBA Season'
```

We then use the yt-dlp library to execute this search and extract the top 10 results. We start from the pragmatic assumption that with a well-constructed query, a relevant video is necessarily found within this top 10.

Next, we apply a series of strict filters to discard parasitic videos that would bias the final figures. Following a principle of parsimony, we prefer to miss a relevant video rather than include irrelevant content. Specifically, we check the broadcaster's identity to ensure they are not on the following exclusion list, which includes official channels and rights holders:

```
["NBA on ESPN", "Sports On Prime", "Prime Video Sport France", "Amazon Prime  
Video Türkiye", "NBA Europe", "NBA G League", "NBA Extra - beIN SPORTS France",  
"NBA", "Atlanta Hawks", "Boston Celtics", "Brooklyn Nets", "Charlotte Hornets",  
"Chicago Bulls", "Cleveland Cavaliers", "Dallas Mavericks", "Denver Nuggets", "Detroit  
Pistons", "Golden State Warriors", "Houston Rockets", "Indiana Pacers", "LA Clippers",  
"Los Angeles Lakers", "Memphis Grizzlies", "Miami Heat", "Milwaukee Bucks",  
"Minnesota Timberwolves", "New Orleans Pelicans", "New York Knicks", "Oklahoma  
City Thunder", "Orlando Magic", "Philadelphia 76ers", "Phoenix Suns", "Portland Trail  
Blazers", "Sacramento Kings", "San Antonio Spurs", "Toronto Raptors", "Utah Jazz",  
"Washington Wizards"]
```

A filter is also applied to the viewership. To avoid wasting time analyzing videos with no impact, we set a minimum threshold of 1,000 views. Preliminary exploration showed that almost all cumulative viewership comes from a few viral videos and that the majority of uploads remain confidential. Discarding these videos therefore does not significantly impact the global volume while saving us precious computation time. This 1,000-view threshold is, moreover, conservative.

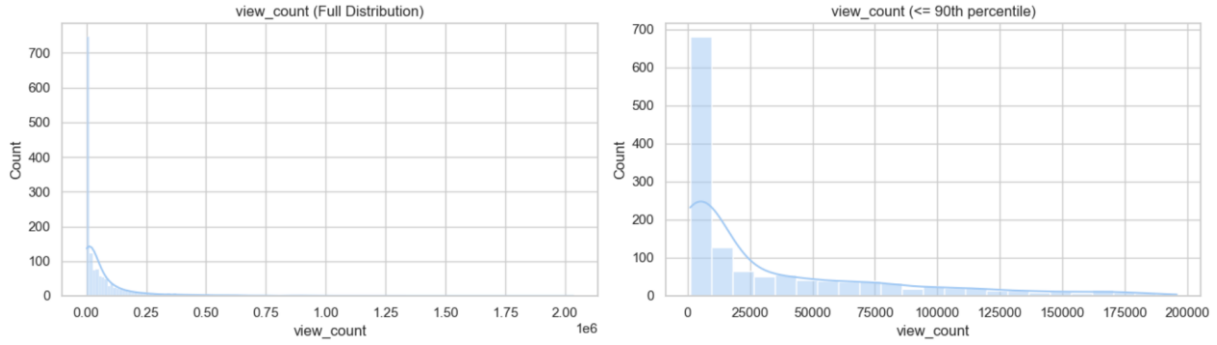


Figure 34: View count distribution for the collected videos.

Next, we analyze the duration. Excessively long videos are removed as they have been identified as NBA 2K video game gameplay, which some users publish while posing as real future games. Finally, we validate the relevance of the content via a multitude of regular expressions (Regex), checking for the presence of the date and team names in the title under various formats (full names, abbreviations, etc.).

To guarantee the relevance of the analyses, we only select matches that are at least one week old. This delay allows for the collection of videos that have reached their full audience potential, thus avoiding the processing of content that is too recent and whose view count would not be representative. This one-week choice is justified by the intensity of the NBA calendar, where each team plays an average of three to four matchups per week. In such a rhythm, spectator interest in the recap of a past game fades quickly in favor of new matchups, making late viewing of highlights marginal after seven days.

Although this threshold considerably limits the bias related to the timing of collection, we still plan for a regular update procedure, ideally monthly. This step is necessary to update the view counts in our database and, by extension, the generated media value. By applying the audience increase percentage to the initial financial calculations, we ensure a dynamic valuation that is faithful to the real evolution of engagement over the long term.

8.2. Justification for the Choice of 5 FPS Sampling

The most critical optimization to enable scaling across thousands of videos lies in the reduction of the temporal sampling frequency. We have fixed this value at **5 frames per second (FPS)**. This downsampling necessarily introduces an approximation compared to the native stream, but it is indispensable for containing the computational load. To scientifically validate this threshold, we conducted a rigorous comparative study by processing a sample of videos at 3, 5, and 10 FPS, using the exhaustive

treatment at 30 FPS as the ground truth. We then measured the deviations on two key metrics: the cumulative exposure time and the average area of the bounding boxes.

The analysis of absolute and relative errors, illustrated by the box plots above, disqualifies the 3 FPS option. As the graphs show, this frequency generates excessive variance, with wide interquartile ranges and numerous outliers, particularly visible in dynamic zones such as the *side-court-logos*. Relative errors frequently exceed 4 to 5% there, introducing a level of statistical noise that is too high for a reliable financial valuation. In contrast, 10 FPS sampling offers near-perfect precision, with relative errors kept below the 1% mark, but provides only a marginal gain compared to the intermediate solution.

The choice of 5 FPS emerges as the optimal balance point. Examination of the relative error graphs reveals a drastic tightening of the error distribution compared to 3 FPS, with the majority of measurements stabilizing within a tolerance range of ± 1 to 2%. The stabilization is particularly notable for the *side-court-led-logo* class, which is highly unstable at low frequencies. The additional precision gain offered by moving to 10 FPS does not justify doubling the calculation time.

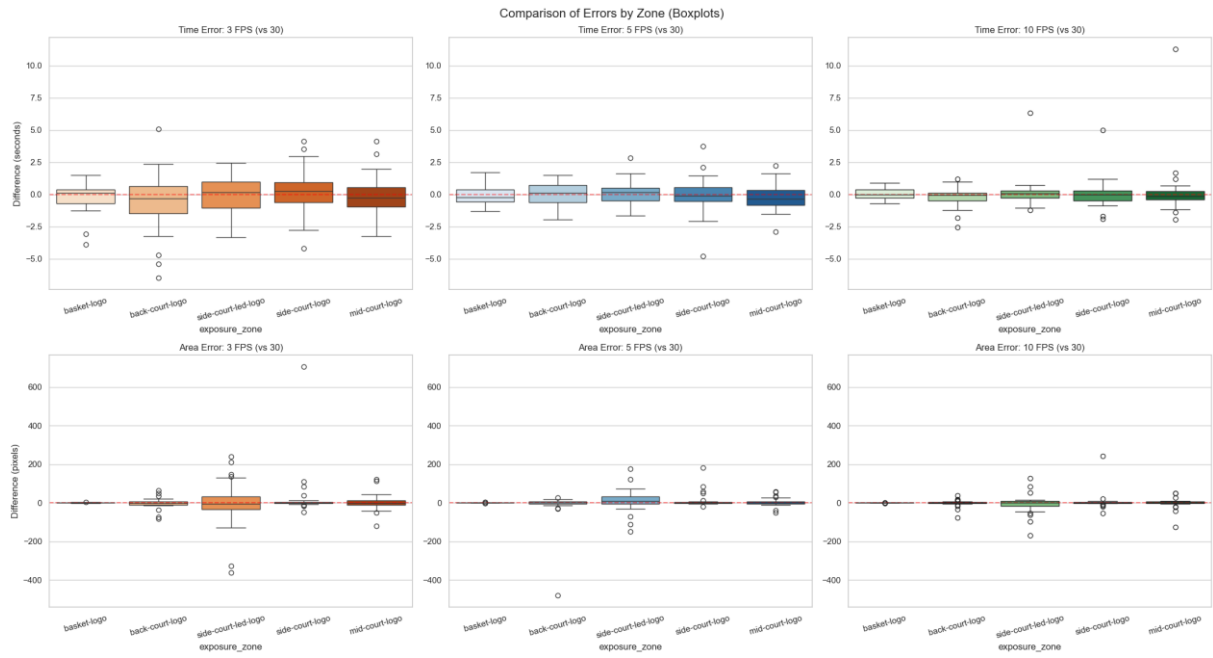


Figure 35: Error analysis of the various approximations by zone.

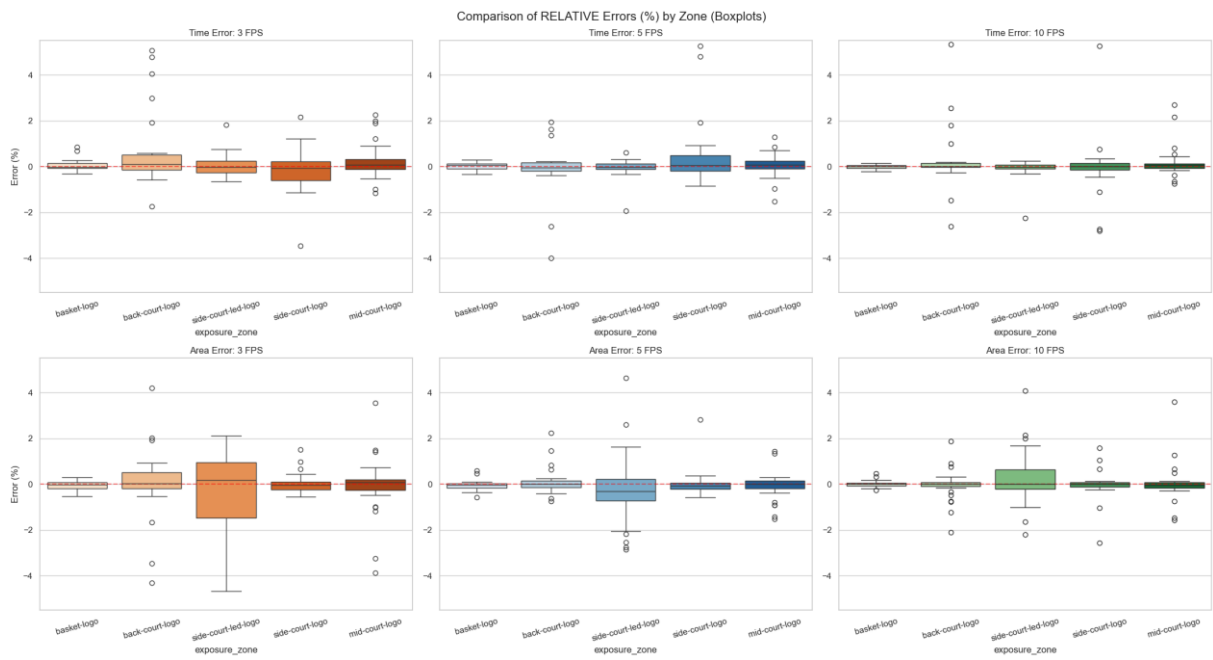


Figure 36: Error percentage analysis of the various approximations by zone.

8.3. Note on the Use of LLMs

In a spirit of transparency, I would like to clarify that I used LLMs (Gemini 3 Pro and Claude 4.5 Sonnet) to assist me with this project. I did not use them to do my thinking for me, but rather as assistants to save time on two specific aspects.

First, I used them for the code. They helped me write my scripts faster and made the code cleaner and easier to read. This was obviously not done at the expense of understanding what was produced. I made sure to fully understand how everything worked to ensure the program did exactly what I wanted.

Second, these tools helped me with my research, particularly with Google Deep Search. This allowed me to quickly build a bibliography on very specific topics (the CPV benchmark, for example) in order to focus my time on the most relevant sources to answer my questions.